dirac16complex: provenance

Sources, decisions, derivations and the verification record of the Dirac equation and the Kohn–Sham theories of the 16-component field dirac16complex — sections 7–10 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states what was asked, what was used (the first part of 7+1.nb and the literature), every decision taken with the equation behind it, how each result was derived, and how it was checked: the 74 checks of sections 7–9 of 7+1.nb with their results, the 90 boolean checks of the independent Python program, the comparison of every printed number, the program versions and the SHA-256 hashes of the files.

Contents

1. The task **2**

2. Inputs **3**
2.1 The notebook 7+1.nb of the first stage . . . 3
2.2 Literature, and the equation taken from each source . . . 4

3. Decisions and derivations **5**
3.1 The real Dirac matrices and the sign of the metric . . . 5
3.2 The Dirac equation, the Klein–Gordon equation and the Hamiltonian . . . 6
3.3 Real plane-wave spinors . . . 6
3.4 Grassmann algebra, quantization and the canonical anticommutators . . . 6
3.5 Charge conjugation . . . 7
3.6 Lorentz covariance and the boost example . . . 7
3.7 Chirality and the two coupled 8-component equations . . . 7
3.8 The self-coupled field . . . 8
3.9 The nonlinear field equation . . . 8
3.10 Hohenberg–Kohn theorem and Kohn–Sham equations . . . 9
3.11 Exchange (Wick) and the Fierz identities . . . 9
3.12 The chiral Kohn–Sham theory . . . 10
3.13 Uniform matter: formulas, parameters and numerical precision . . . 10
3.14 The chiral vacuum with the Dirac sea . . . 11
3.15 Notation decisions . . . 11
3.16 The signature: the author's explanation of 27 October 2026 . . . 12

4. Verification record **12**
4.1 The Mathematica run . . . 12
4.2 The checks of sections 7–9 of 7+1.nb and their results . . . 13
4.3 Independent Python checks: tools/ref_dirac16.py . . . 18
4.4 Hashes of the files . . . 24

5. Process and environment **25**

6. Authorship **26**

1. The task

This stage of the project asked for the following (summarized; the original wording is the author's private working material and is not reproduced):

1. Back up 7+1.nb, then append to it (the first part of the notebook stays as it is).
2. A valid and checked Dirac equation on the real 8-dimensional spacetime with one time and seven space dimensions, for a 16-component quantized fermion field named dirac16complex whose components are complex-valued Grassmann (anticommuting) operators.
3. Eight 16×16 Dirac matrices taken from an appropriate set of anticommuting real matrices of 7+1.nb; they MUST be purely real.
4. Chiral projections that write the 16-component field equation as two 8-component field equations, coupled when the mass is not zero.
5. A 16-component Kohn–Sham theory of dirac16complex coupled to itself.
6. A chiral version of that Kohn–Sham theory, using the chiral projections.
7. Complete, precise and checked documentation; for every documentation file a companion provenance file and a symbol file, each in Markdown and LaTeX, the LaTeX compiled to PDF; no document may refer the reader to another document, and every equation used must be written in the document that uses it.
8. Push every completed stage to the repository github.com/once-ere/7plus1 (the repository was renamed from 7+1 to 7plus1 during this stage), respecting the repository's .gitignore.
9. Use only the Wolfram front end WolframNB.exe of Wolfram 15.0.1, installed in `C:\Program Files\Wolfram Research\Wolfram\15.0.1`.

The coordinates are those of 7+1.nb and were not renamed: $X = (x_0, x_1, \ldots, x_7)$ with x_0 the ordinary time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary 3-space and $x_4, \ldots, x_7$ four auxiliary space-like dimensions. How each requirement was met:

#	how it was met	where in 7+1.nb
1	the evaluated notebook of the first stage was copied to backups/7+1_2026-10-26_before_dirac16complex.nb (section 2.1); sections 0–6 of the new notebook are cell for cell identical to it (section 4.3), except for the cells of the signature explanation of 27 October 2026 (section 3.16)	sections 0–6 unchanged (but for 3.16)
2	$(i\Gamma^A\partial_A - m)\psi = 0$ for `dirac16complex = Table[ψ16[a] @@ X, {a, 1, 16}]`; Klein–Gordon, Hamiltonian, plane waves, Grassmann algebra, mode expansion, canonical anticommutators, charge conjugation and Lorentz covariance (section 3.2–3.6)	7.1–7.9
3	$\Gamma_0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i = \left(\begin{smallmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{smallmatrix}\right)$: the real generators `Γ17[A]` of the first stage; every entry 0 or ±1 (section 3.1)	7.1
4	complex chirality $\chi = i\Gamma_0 \cdots \Gamma_7$ (no real one exists), $P_{L,R} = \frac{1}{2}(1 \mp \chi)$, and the two coupled equations $(i\partial_0 \pm \tau_i^O \partial_i)\phi_{L,R} = m\,\phi_{R,L}$ (section 3.7)	7.10–7.11
5	self-coupling through scalar, pseudoscalar, vector and axial-vector contact interactions; Hohenberg–Kohn theorem; Kohn–Sham equations; exchange (Wick), Fierz; uniform matter (sections 3.8–3.11, 3.13)	8.1–8.7
6	chiral densities κ, j_L, j_R; complex Kohn–Sham mass; 8+8-component Kohn–Sham equations; chiral symmetry; the chiral vacuum with the Dirac sea (sections 3.12, 3.14)	9.1–9.6
7	the documentation of this stage (a main document, a symbol dictionary and this record, each .md/.tex/.pdf), generated by tools/make_docs.py (section 5)	—
8	commits pushed to github.com/once-ere/7plus1 by the Wolfram job `tools/git_push.wl` (section 5)	—
9	every evaluation of 7+1.nb was made by that front end (section 4.1)	—

2. Inputs

2.1 The notebook 7+1.nb of the first stage

Before anything was appended, the evaluated notebook of the first stage (128 checks, all True) was copied, in the workspace `D:\Developer\github\7+1`, to the file below. The folder backups/ is excluded from the repository by its .gitignore, so the copy stays in the workspace only.

```
backups/7+1_2026-10-26_before_dirac16complex.nb
306963 bytes
SHA-256 f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce
```

Sections 7–10 use the following objects of sections 1–5 of 7+1.nb, unchanged (Mathematica names in typewriter). Coordinates, metric and derivative operators:

$X = (x_0, x_1, \ldots, x_7)$ (`X`), $\quad \eta = \mathrm{diag}(+1,-1,-1,-1,-1,-1,-1,-1)$ (`η17`), $\quad \partial_A = \partial/\partial x_A$,
$\Gamma^A \partial_A \psi$ (`DiracOp17`), $\quad \Box\psi = \eta^{AB}\partial_A\partial_B\psi = \partial_0^2\psi - \partial_1^2\psi - \cdots - \partial_7^2\psi$ (`waveOp17`).

The real 8×8 octonion matrices `τO[i]` ($\tau_0^O = 1_8$), obtained in the first stage from the split-octonion matrices τ_A of 4+4.nb as $\tau_i^O = \tau_i$ $(i = 1,2,3)$ and $\tau_k^O = \tau_k\,\sigma$ $(k = 4,\ldots,7)$, $\sigma = \tau_1\tau_2\tau_3 = \left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$:

$$\tau_1^O = \begin{pmatrix} 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_2^O = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_3^O = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_4^O = \begin{pmatrix} 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0 \end{pmatrix},$$

$$\tau_5^O = \begin{pmatrix} 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0 \end{pmatrix}, \quad \tau_6^O = \begin{pmatrix} 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0 \end{pmatrix},$$

$$\tau_7^O = \begin{pmatrix} 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0 \end{pmatrix};$$

the split matrices used for them are

$$\tau_4 = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_5 = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_6 = \begin{pmatrix} 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_7 = \begin{pmatrix} -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1 \end{pmatrix},$$

and $\tau_i = \tau_i^O$ for $i = 1, 2, 3$. They satisfy

$$(\tau_i^O)^{\mathsf{T}} = -\tau_i^O, \qquad \tau_i^O \tau_j^O + \tau_j^O \tau_i^O = -2\,\delta_{ij}\,1_8, \qquad \tau_1^O \cdots \tau_7^O = 1_8.$$

The real 16×16 generators `Γ17[A]` of the Clifford algebra of 7+1 spacetime, their products and the objects built from them:

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}, \qquad \Gamma^A = \eta^{AA}\Gamma_A \ (\texttt{Γ17up[A]}),$$

$$\Gamma_8 \equiv \omega = \Gamma_0\Gamma_1\cdots\Gamma_7 \ (\texttt{Γ17[8]}), \qquad \chi = i\,\omega \ (\texttt{χ17}), \qquad P_{L,R} = \tfrac{1}{2}(1_{16} \mp \chi) \ (\texttt{PL17},\ \texttt{PR17}),$$

$$\Sigma_{AB} = \tfrac{1}{4}(\Gamma_A\Gamma_B - \Gamma_B\Gamma_A) \ (\texttt{Σ17[A, B]}), \qquad \Gamma_K = \Gamma_{A_1}\cdots\Gamma_{A_k} \ (K = \{A_1 < \cdots < A_k\};\ \texttt{products256}),$$

$$\mathcal{B}(\beta_{\mathrm{b}}) = \cosh\tfrac{\beta_{\mathrm{b}}}{2}\,1_{16} + \sinh\tfrac{\beta_{\mathrm{b}}}{2}\,\Gamma_0\Gamma_1 \ (\texttt{Sβ}), \qquad L_{BK} = \tfrac{1}{16}\mathrm{tr}\big(\eta_{BB}\Gamma_B\,\mathcal{B}\,\Gamma_K\,\mathcal{B}^{-1}\big) \ (\texttt{Λboost}).$$

In terms of the 16×16 Dirac matrices of 4+4.nb, $T_A = \left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$ with $\bar\tau_0 = 1_8$, $\bar\tau_A = \sigma\tau_A^{\mathsf{T}}\sigma$ and $T_AT_B + T_BT_A = 2\eta_{AB}^{(4,4)}1_{16}$, $\eta^{(4,4)} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$, the first stage obtained $\Gamma_0 = T_0$, $\Gamma_i = T_0T_i$ $(i = 1, 2, 3)$ and $\Gamma_k = T_0T_kT_4T_5T_6T_7$ $(k = 4, \ldots, 7)$.

2.2 Literature, and the equation taken from each source

- P. A. M. Dirac, Proc. R. Soc. Lond. A 117 (1928) 610 — the first-order wave equation $(i\gamma^\mu\partial_\mu - m)\psi = 0$ with $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$; here written in 7+1 dimensions with the eight Γ^A.
- P. Hohenberg and W. Kohn, Phys. Rev. 136 (1964) B864 — for a non-degenerate ground state the density determines the external potential up to a constant, and $E[v] = \min_n \big(F[n] + \int v\,n\big)$; the proof by the two variational inequalities is used in section 3.10.
- W. Kohn and L. J. Sham, Phys. Rev. 140 (1965) A1133 — a non-interacting reference system with the same density: $\big[-\tfrac{1}{2}\nabla^2 + v + v_H + v_{xc}\big]\varphi_k = \varepsilon_k\varphi_k$, $n = \sum_k |\varphi_k|^2$, $v_{xc} = \delta E_{xc}/\delta n$ (atomic units); here its relativistic form.
- A. K. Rajagopal and J. Callaway, Phys. Rev. B 7 (1973) 1912; A. H. MacDonald and S. H. Vosko, J. Phys. C 12 (1979) 2977 — relativistic density-functional theory: the energy is a functional of the current j^A, and the Kohn–Sham equations have the Dirac form $\big[\alpha_i(-i\partial_i + \mathcal{V}_i) + \beta m + \mathcal{V}_0\big]\varphi_k = \varepsilon_k\varphi_k$ with the effective potential $\mathcal{V}_A = \delta(E - T_s)/\delta j^A$.
- M. Levy, Proc. Natl. Acad. Sci. USA 76 (1979) 6062; E. H. Lieb, Int. J. Quantum Chem. 24 (1983) 243 — the universal functional by constrained search, $F[n] = \min_{\Psi\to n}\langle\Psi|\hat T + \hat W|\Psi\rangle$, and Lieb's form $F[n] = \sup_v \big(E[v] - \int v\,n\big)$.

- R. M. Dreizler and E. K. U. Gross, *Density Functional Theory*, Springer (1990), ch. 8; E. Engel and R. M. Dreizler, *Density Functional Theory: An Advanced Course*, Springer (2011) — the no-sea approximation: the densities are sums over the occupied positive-energy Kohn–Sham states only, $j^A = \sum_{0<\varepsilon_k\le\varepsilon_F} \bar\varphi_k \Gamma^A \varphi_k$.
- J. D. Walecka, Ann. Phys. 83 (1974) 491; B. D. Serot and J. D. Walecka, Adv. Nucl. Phys. 16 (1986) 1 — relativistic mean field of uniform matter: $m^* = m - \frac{g_\sigma^2}{m_\sigma^2}\rho_S$, $V_0 = \frac{g_\omega^2}{m_\omega^2}\rho$, $\varepsilon = \frac{g_\omega^2}{2m_\omega^2}\rho^2 + \frac{m_\sigma^2}{2g_\sigma^2}(m - m^*)^2 + \varepsilon_{\rm kin}(m^*)$; the contact couplings here correspond to $g_S = -g_\sigma^2/m_\sigma^2$ and $g_V = g_\omega^2/m_\omega^2$.
- B. A. Nikolaus, T. Hoch and D. G. Madland, Phys. Rev. C 46 (1992) 1757 — relativistic point-coupling (contact) models, with four-fermion terms of the form $-\frac{\alpha_S}{2}(\bar\psi\psi)^2 - \frac{\alpha_V}{2}(\bar\psi\gamma_\mu\psi)(\bar\psi\gamma^\mu\psi)$, the 3+1-dimensional analogue of the g_S and g_V terms here.
- N. M. Hugenholtz and L. Van Hove, Physica 24 (1958) 363 — at saturation (zero pressure $P = \rho\mu - \varepsilon$) the Fermi energy equals the energy per particle, $\varepsilon_F = \mu = E/A$.
- G. C. Wick, Phys. Rev. 80 (1950) 268 — expectation values in a Slater determinant factorize into contractions: $\langle c_a^\dagger c_c^\dagger c_d c_b\rangle = \varrho_{ba}\varrho_{dc} - \varrho_{da}\varrho_{bc}$, $\varrho_{xy} = \langle c_y^\dagger c_x\rangle$.
- M. Fierz, Z. Phys. 104 (1937) 553 — rearrangement of four-fermion products from the completeness of the Clifford basis, here $\sum_K (\Gamma_K)_{ab}(\Gamma_K^{-1})_{cd} = 16\,\delta_{ad}\delta_{cb}$.
- Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122 (1961) 345; S. P. Klevansky, Rev. Mod. Phys. 64 (1992) 649 — chirally symmetric contact coupling $G[(\bar\psi\psi)^2 + (\bar\psi i\gamma_5\psi)^2]$ with a momentum cutoff; gap equation $M = m - 2G\langle\bar\psi\psi\rangle$; for $m = 0$ a solution $M \neq 0$ exists above a critical coupling (here $G = -g/2$).
- F. A. Berezin, *The Method of Second Quantization*, Academic Press (1966) — Grassmann algebra, $\theta_k\theta_l = -\theta_l\theta_k$, and the left derivative.
- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton (1989) — the classification of real Clifford algebras. In the convention used here ($Cl(p,q)$ has p generators squaring to $+1$ and q to -1) it gives $Cl(1,7) \cong \mathbb{R}(16)$ and $Cl(7,1) \cong \mathbb{H}(8)$; Lawson and Michelsohn define $v^2 = -q(v)$, so their $Cl_{7,1}$ is the $Cl(1,7)$ of this record.

3. Decisions and derivations

Each subsection states a decision or a derived result, the equations behind it, and the checks of 7+1.nb that confirm it (quoted by their names in section 4.2).

3.1 The real Dirac matrices and the sign of the metric

Decision. The Dirac matrices are the real generators Γ_A of section 2.1, with

$$\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}, \qquad \eta = \mathrm{diag}(+1,-1,-1,-1,-1,-1,-1,-1): \qquad \Gamma_0^2 = 1_{16}, \quad \Gamma_i^2 = -1_{16}.$$

Reason. The task requires real matrices, and only this overall sign allows them; it is the sign pattern of the octonion basis, η being simply the squares $e_0^2 = +1$, $e_i^2 = -1$ of the octonion units. With the opposite sign $\eta' = -\eta$ the relations $R_AR_B + R_BR_A = 2\eta'_{AB}1_{16}$ define $Cl(7,1) \cong \mathbb{H}(8)$. If real 16×16 matrices R_A satisfied them, then, because $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ has only one irreducible module, $R_A = T(i\Gamma_A)T^{-1}$ for an invertible complex T ($i\Gamma_A$ satisfy the same relations). $R_A = \overline{R_A}$ gives $Y\overline{(i\Gamma_A)} = (i\Gamma_A)Y$ with $Y = T^{-1}\overline{T}$, $Y\overline{Y} = 1_{16}$; since $\overline{(i\Gamma_A)} = -i\Gamma_A$ this is $Y\Gamma_A = -\Gamma_A Y$, so $Y\omega$ commutes with all Γ_A and $Y = c\,\omega$ (Schur), whence $Y\overline{Y} = |c|^2\omega^2 = -|c|^2 1_{16} \neq 1_{16}$. Renaming, reordering or re-signing the indices does not change the signature of the relation (the numbers of generators squaring to $+1$ and to -1). The first stage proved this in 7+1.nb.

The Dirac matrices with upper index, α, β, ω (checks 1–5 of section 7.1 of 7+1.nb):

$$\Gamma^A = \eta^{AA}\Gamma_A, \qquad (\Gamma^A)^\dagger = \Gamma^0\Gamma^A\Gamma^0, \qquad \beta = \Gamma^0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix},$$

$$\alpha_i = \Gamma^0\Gamma^i = \begin{pmatrix} 0 & \tau_i^O \\ -\tau_i^O & 0 \end{pmatrix}, \qquad \alpha_i\alpha_j + \alpha_j\alpha_i = 2\delta_{ij}1_{16}, \qquad \alpha_i\beta + \beta\alpha_i = 0, \qquad \beta^2 = 1_{16},$$
$$\omega = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \omega^2 = -1_{16}, \qquad \omega\Gamma_A = -\Gamma_A\omega.$$

3.2 The Dirac equation, the Klein–Gordon equation and the Hamiltonian

Decision. $(i\Gamma^A\partial_A - m)\psi = 0$. The factor i is unavoidable: with real Γ_A, $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A,\Gamma^B\}\partial_A\partial_B = \Box$, so

$$\big(i\Gamma^A\partial_A + m\big)\big(i\Gamma^A\partial_A - m\big) = -\big(\Box + m^2\big)1_{16},$$

which is the Klein–Gordon operator with the correct sign for a massive particle; without the i the mass term would have the wrong sign ($(\Gamma^A\partial_A)^2 - m^2 = \Box - m^2$). Multiplying by Γ^0:

$$\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 - h\big)\psi, \qquad h = -i\alpha_i\partial_i + \beta m, \qquad h^\dagger = h,$$

because α_i, β are real symmetric and $-i\partial_i$ is Hermitian. Conservation of $j^A = \bar\psi\Gamma^A\psi$, $\bar\psi = \psi^\dagger\Gamma^0$, follows from the adjoint equation $-i\partial_A\bar\psi\Gamma^A - m\bar\psi = 0$: $\partial_A j^A = (im\bar\psi)\psi + \bar\psi(-im\psi) = 0$. 7+1.nb checks these identities for 16 arbitrary functions $\psi_a(x_0,\dots,x_7)$.

3.3 Real plane-wave spinors

For $e^{-i(Ex_0 - \vec p\cdot\vec x)}$, h becomes the real symmetric matrix

$$h(\vec p) = p^i\alpha_i + m\beta = \begin{pmatrix} 0 & M \\ M^{\mathsf T} & 0 \end{pmatrix}, \qquad M = m1_8 + p^i\tau_i^O,$$
$$MM^{\mathsf T} = m^2 1_8 - (p^i\tau_i^O)^2 = E^2 1_8, \qquad E = \sqrt{m^2 + \vec p^{\,2}},$$

using $(p^i\tau_i^O)^{\mathsf T} = -p^i\tau_i^O$ and $(p^i\tau_i^O)^2 = -\vec p^{\,2}1_8$. **Derivation of the spinors.** With $u_s = \frac{1}{\sqrt E}(E\,e_s;\ M^{\mathsf T}e_s)$ and $v_s = \frac{1}{\sqrt E}(E\,e_s;\ -Me_s)$, and $h(-\vec p) = \left(\begin{smallmatrix} 0 & M^{\mathsf T} \\ M & 0 \end{smallmatrix}\right)$:

$$h(\vec p)\,(E\,e_s;\ M^{\mathsf T}e_s) = (MM^{\mathsf T}e_s;\ E\,M^{\mathsf T}e_s) = E\,(E\,e_s;\ M^{\mathsf T}e_s),$$
$$h(-\vec p)\,(E\,e_s;\ -Me_s) = (-M^{\mathsf T}Me_s;\ E\,Me_s) = -E\,(E\,e_s;\ -Me_s),$$
$$u_s^{\mathsf T}u_r = \frac{E^2\delta_{sr} + e_s^{\mathsf T}MM^{\mathsf T}e_r}{E} = 2E\,\delta_{sr}, \qquad v_s^{\mathsf T}v_r = 2E\,\delta_{sr}.$$

Decision (normalization). $u^{\mathsf T}u = 2E$, the usual covariant normalization, so that $\sum_s u_s\bar u_s = \not p + m$ and $\sum_s v_s\bar v_s = \not p - m$ with $\not p = E\Gamma_0 + p^i\Gamma_i$. The example $\vec p = (2,2,4,0,0,0,0)$, $m = 1$, $E = 5$ was chosen because E is an integer, so $\sqrt5\,u_s$ has integer entries. The real orthogonal matrix $U = \frac{1}{\sqrt{2E}}\big(u_1(\vec p),\dots,u_8(\vec p),v_1(-\vec p),\dots,v_8(-\vec p)\big)$ satisfies $U^{\mathsf T}h(\vec p)U = \mathrm{diag}(E1_8, -E1_8)$, equivalently the completeness relation

$$\sum_{s=1}^{8} u_s(\vec p)\,u_s(\vec p)^{\mathsf T} + \sum_{s=1}^{8} v_s(-\vec p)\,v_s(-\vec p)^{\mathsf T} = 2E\,1_{16}.$$

3.4 Grassmann algebra, quantization and the canonical anticommutators

Decision. The Grassmann algebra is implemented with ordered monomials `grass[{k1, …, kr}]`, the product $\theta_K\theta_{K'} = \mathrm{sign}(\pi)\theta_{\mathrm{sort}(K\cup K')}$ (zero if $K\cap K' \neq \emptyset$) and the left derivative $\partial^L(\theta_{k_1}\cdots\theta_{k_r})/\partial\theta_{k_j} = (-1)^{j-1}\theta_{k_1}\cdots\widehat{\theta_{k_j}}\cdots\theta_{k_r}$, as in Berezin. The current-conservation check uses $\psi = \sum_{k=1}^{3}\theta_k U_k$, $\bar\psi = \sum_k \theta_{3+k}U_k^\dagger\Gamma^0$ with

$$U_1 = u_1(\vec p_1)e^{-i(5x_0 - \vec p_1\cdot\vec x)}, \qquad U_2 = v_3(\vec p_3)e^{+i(3x_0 - \vec p_3\cdot\vec x)}, \qquad U_3 = u_5(\vec 0)e^{-ix_0},$$
$$\vec p_1 = (2,2,4,0,0,0,0), \qquad \vec p_3 = (0,0,0,2,2,0,0).$$

The mode expansion and the anticommutators of the modes

$$\psi(x_0,\vec{x}) = \sum_{\vec{p}}\sum_{s=1}^{8} \frac{1}{\sqrt{2EV}}\Big[b_s(\vec{p})u_s(\vec{p})e^{-i(Ex_0-\vec{p}\cdot\vec{x})} + d_s^\dagger(\vec{p})v_s(\vec{p})e^{+i(Ex_0-\vec{p}\cdot\vec{x})}\Big],$$
$$\{b_s(\vec{p}), b_r^\dagger(\vec{q})\} = \{d_s(\vec{p}), d_r^\dagger(\vec{q})\} = \delta_{sr}\delta_{\vec{p}\vec{q}} \quad \text{(others 0)}$$

give, with the completeness relation of section 3.3 (the v term contributes at $-\vec{p}$, which the symmetric sum over $\vec{p}$ relabels),

$$\{\psi_a(x_0,\vec{x}), \psi_b^\dagger(x_0,\vec{y})\} = \frac{1}{V}\sum_{\vec{p}} \frac{e^{i\vec{p}\cdot(\vec{x}-\vec{y})}}{2E}\Big[\sum_s u_s u_s^{\mathsf{T}} + \sum_s v_s(-\vec{p})v_s(-\vec{p})^{\mathsf{T}}\Big]_{ab}$$
$$= \delta_{ab}\frac{1}{V}\sum_{\vec{p}} e^{i\vec{p}\cdot(\vec{x}-\vec{y})} = \delta_{ab}\,\delta^7(\vec{x}-\vec{y}).$$

Decision (test set). The exact check uses five momenta closed under $\vec{p}\to-\vec{p}$, $\pm(2,2,4,0,0,0,0)$, $\pm(0,0,0,2,2,0,0)$, $\vec{0}$ ($m=1$, $E=5,5,3,3,1$, all integers), with symbolic $\vec{x}$, $\vec{y}$ and box volume V. The normal-ordered Hamiltonian and charge follow from $U^{\mathsf{T}}hU = \mathrm{diag}(E,-E)$: $H=\sum E(b^\dagger b + d^\dagger d)$, $Q=\sum(b^\dagger b - d^\dagger d)$.

3.5 Charge conjugation

Decision. $\psi^c = \omega\psi^*$. Because the Γ_A are real, conjugating the Dirac equation gives $(-i\Gamma^A\partial_A - m)\psi^* = 0$, and ω is real and anticommutes with every Γ_A:

$$\big(i\Gamma^A\partial_A - m\big)\omega\psi^* = \omega\big(-i\Gamma^A\partial_A - m\big)\psi^* = 0, \qquad (\psi^c)^c = \omega\omega\psi = -\psi, \qquad (\not{p}+m)\,\omega\,u_s(\vec{p}) = 0.$$

Consequently $\psi^c=\psi$ has no non-zero solution: a massive real (Majorana) field needs 32 real components in 7+1 dimensions.

3.6 Lorentz covariance and the boost example

The spinor boost $\mathcal{B}(\beta_{\mathrm{b}}) = e^{\beta_{\mathrm{b}}\Sigma_{01}}$ and its Lorentz matrix L (section 2.1) satisfy $\mathcal{B}\Gamma_K\mathcal{B}^{-1} = \sum_B L_{BK}\Gamma_B$ and $L^{\mathsf{T}}\eta L = \eta$, with

$$L = \begin{pmatrix} \cosh\beta_{\mathrm{b}} & -\sinh\beta_{\mathrm{b}} & 0 \\ -\sinh\beta_{\mathrm{b}} & \cosh\beta_{\mathrm{b}} & 0 \\ 0 & 0 & 1_6 \end{pmatrix}.$$

Decision (example). $\beta_{\mathrm{b}} = 2\log 2$, because then every number is rational: $\cosh\frac{\beta_{\mathrm{b}}}{2} = \frac{5}{4}$, $\sinh\frac{\beta_{\mathrm{b}}}{2} = \frac{3}{4}$, $\cosh\beta_{\mathrm{b}} = \frac{17}{8}$, $\sinh\beta_{\mathrm{b}} = \frac{15}{8}$. For $p=(5,2,2,4,0,0,0,0)$, $m=1$:

$$p' = Lp = \Big(\tfrac{17}{8}\cdot5 - \tfrac{15}{8}\cdot2,\ -\tfrac{15}{8}\cdot5 + \tfrac{17}{8}\cdot2,\ 2,4,0,0,0,0\Big) = \Big(\tfrac{55}{8}, -\tfrac{41}{8}, 2,4,0,0,0,0\Big), \qquad p'\cdot p' = \tfrac{3025-1681}{64} - 20 = 1,$$

and $(\not{p}' - m)\,\mathcal{B}\,u_s(\vec{p}) = 0$ for $s=1,\dots,8$, with $\mathcal{B} = \frac{5}{4}\,1_{16} + \frac{3}{4}\,\Gamma_0\Gamma_1$.

3.7 Chirality and the two coupled 8-component equations

Result. No real chirality matrix exists: the real 16×16 matrices Z with $Z\Gamma_A = -\Gamma_A Z$ ($A=0,\dots,7$) form the space $\mathbb{R}\,\omega$ (7+1.nb solves the 256-unknown linear system, rank 255), and $(c\,\omega)^2 = -c^2 1_{16}$ cannot be 1_{16}. **Decision.** Chirality is the Hermitian matrix $\chi = i\omega$, unique up to sign; the sign is chosen so that $P_L = \frac{1}{2}(1_{16}-\chi)$ projects on $\chi=-1$:

$$\chi = \begin{pmatrix} 0 & -i1_8 \\ i1_8 & 0 \end{pmatrix}, \qquad \chi^2 = 1_{16}, \qquad \chi\Gamma_A = -\Gamma_A\chi, \qquad \chi\Sigma_{AB} = \Sigma_{AB}\chi,$$
$$E_L = \frac{1}{\sqrt{2}}\begin{pmatrix} 1_8 \\ -i1_8 \end{pmatrix}, \quad E_R = \frac{1}{\sqrt{2}}\begin{pmatrix} -i1_8 \\ 1_8 \end{pmatrix}, \qquad \chi E_L = -E_L, \quad \chi E_R = E_R, \quad E_L E_L^\dagger = P_L, \quad E_R E_R^\dagger = P_R.$$

Decision (chiral components). $\phi_L = E_L^\dagger\psi = (\psi_\uparrow + i\psi_\downarrow)/\sqrt{2}$ and $\phi_R = E_R^\dagger\psi = (i\psi_\uparrow + \psi_\downarrow)/\sqrt{2}$. These isometries were chosen because they make the chiral blocks of the Dirac matrices as simple as possible:

$$E_L^\dagger\beta E_R = E_R^\dagger\beta E_L = 1_8, \qquad E_L^\dagger\beta E_L = E_R^\dagger\beta E_R = 0,$$
$$E_L^\dagger\alpha_i E_L = -\sigma_i, \qquad E_R^\dagger\alpha_i E_R = \sigma_i, \qquad E_L^\dagger\alpha_i E_R = 0, \qquad \sigma_i = i\tau_i^O.$$

Derivation of the chiral equations. Insert $\psi = E_L\phi_L + E_R\phi_R$ into $\Gamma^0(i\Gamma^A\partial_A - m)\psi = (i\partial_0 + i\alpha_i\partial_i - \beta m)\psi$ and project with $E_L^\dagger$, $E_R^\dagger$; since $i\,(-\sigma_i)\partial_i = \tau_i^O\partial_i$:

$$E_L^\dagger\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 + \tau_i^O\partial_i\big)\phi_L - m\,\phi_R,$$
$$E_R^\dagger\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 - \tau_i^O\partial_i\big)\phi_R - m\,\phi_L.$$

So the Dirac equation is equivalent to $(i\partial_0 + \tau_i^O\partial_i)\phi_L = m\phi_R$ and $(i\partial_0 - \tau_i^O\partial_i)\phi_R = m\phi_L$, coupled exactly when $m \neq 0$, with only real 8×8 matrices besides the i of $i\partial_0$. Each obeys the Klein–Gordon equation because $(i\partial_0 - \tau_i^O\partial_i)(i\partial_0 + \tau_j^O\partial_j) = -\Box 1_8$. For a plane wave the equations read $(E + \sigma\cdot\vec{p})\phi_L = m\phi_R$, $(E - \sigma\cdot\vec{p})\phi_R = m\phi_L$, and $W = (E_L, E_R)$ gives $W^\dagger h(\vec{p})W = \left(\begin{smallmatrix} -\sigma\cdot\vec{p} & m1_8 \\ m1_8 & \sigma\cdot\vec{p}\end{smallmatrix}\right)$.

3.8 The self-coupled field

Decision (interaction). The field is coupled to itself by the four Lorentz-invariant contact (zero-range) interactions built from bilinears without derivatives that are used in relativistic point-coupling models: scalar, pseudoscalar, vector and axial vector,

$$\mathcal{L} = \bar\psi\big(i\Gamma^A\partial_A - m\big)\psi - \bar\psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi$$
$$- \frac{g_S}{2}(\bar\psi\psi)^2 - \frac{g_P}{2}(\bar\psi\, i\chi\psi)^2 - \frac{g_V}{2}(\bar\psi\Gamma^A\psi)(\bar\psi\Gamma_A\psi) - \frac{g_A}{2}(\bar\psi\Gamma^A\chi\psi)(\bar\psi\Gamma_A\chi\psi),$$

with real external sources $\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A$ that define the densities. The tensor channel $(\bar\psi\Sigma_{AB}\psi)^2$ and the channels of higher rank are not included; they could be added in the same way, and by the Fierz identities of section 3.11 their exchange terms are again combinations of direct terms. The four densities are real because $\Gamma^0 O$ is Hermitian for $O = 1_{16},\ i\chi,\ \Gamma^A,\ \Gamma^A\chi$. **Units.** $\hbar = c = 1$; ψ has mass dimension 7/2 and every g mass dimension -6, so the theory is an effective one and needs either the no-sea approximation (section 3.10) or a momentum cutoff (section 3.14). **Signs.** The interaction has no time derivatives, so it enters the Hamiltonian density with the opposite sign:

$$\mathcal{H} = :\psi^\dagger(-i\alpha_i\partial_i + \beta m)\psi: + :\bar\psi\big(\mathsf{s} + i\chi\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\mathsf{a}_A)\big)\psi:$$
$$+ \frac{g_S}{2}:\rho_S^2: + \frac{g_P}{2}:\rho_P^2: + \frac{g_V}{2}:j^A j_A: + \frac{g_A}{2}:j_5^A j_{5A}:,$$

with $\rho_S = \bar\psi\psi$, $\rho_P = \bar\psi i\chi\psi$, $j^A = \bar\psi\Gamma^A\psi$, $j_5^A = \bar\psi\Gamma^A\chi\psi$; $g_S < 0$ attracts.

3.9 The nonlinear field equation

The Euler–Lagrange equation is taken with the Grassmann LEFT derivative with respect to $\bar\psi_a$; the bilinears are Grassmann-even, so

$$\big(i\Gamma^A\partial_A - m - \mathsf{s} - i\chi\mathsf{p} - \Gamma^A(\mathsf{v}_A + \chi\mathsf{a}_A)\big)\psi$$
$$= g_S(\bar\psi\psi)\psi + g_P(\bar\psi i\chi\psi)\, i\chi\psi + g_V(\bar\psi\Gamma_A\psi)\Gamma^A\psi + g_A(\bar\psi\Gamma_A\chi\psi)\Gamma^A\chi\psi.$$

Decision (check). 7+1.nb verifies $\partial^L\mathcal{L}_{\rm int}/\partial\bar\psi_a$ for all 16 components in a Grassmann algebra with the 32 generators $\psi_a = \theta_a$, $\bar\psi_a = \theta_{16+a}$, with symbolic couplings.

3.10 Hohenberg–Kohn theorem and Kohn–Sham equations

With $n = (\rho_S, \rho_P, j^A, j_5^A)$ (18 densities), $w = (\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A)$ and $H[w] = H_0 + H_{\rm int} + \int w \cdot n \, d^7x$ in the sector of fixed charge $Q = N$, two sources $w \neq w'$ with different non-degenerate ground states Ψ, Ψ' and the same densities would give

$$E < \langle \Psi' | H[w] | \Psi' \rangle = E' + \int (w - w') \cdot n, \qquad E' < E + \int (w' - w) \cdot n \qquad \Rightarrow \qquad E + E' < E + E',$$

a contradiction; so $E[w] = \min_n \big(F[n] + \int w \cdot n \big)$ with $F[n] = \min_{\Psi \to n} \langle \Psi | H_0 + H_{\rm int} | \Psi \rangle$. Writing $F = T_s + E_H + E_{xc}$ with

$$T_s = \sum_{k=1}^{N} \int \varphi_k^\dagger (-i\alpha_i \partial_i + \beta m) \varphi_k, \qquad E_H = \int \Big[\frac{g_S}{2} \rho_S^2 + \frac{g_P}{2} \rho_P^2 + \frac{g_V}{2} j^A j_A + \frac{g_A}{2} j_5^A j_{5A} \Big],$$

and varying $E = T_s + \int w \cdot n + E_H + E_{xc}$ with respect to $\varphi_k^\dagger$ under $\int \varphi_k^\dagger \varphi_l = \delta_{kl}$ (multipliers ε_k) gives $h_{\rm KS} \varphi_k = \varepsilon_k \varphi_k$ with

$$h_{\rm KS} = \alpha_i (-i\partial_i + \mathcal{V}_i) + \beta (m + \mathcal{S}) + \mathcal{V}_0 + \beta i \chi \mathcal{P} + \chi \mathcal{A}_0 + \alpha_i \chi \mathcal{A}_i,$$
$$\mathcal{S} = \mathsf{s} + g_S \rho_S + \frac{\delta E_{xc}}{\delta \rho_S}, \quad \mathcal{P} = \mathsf{p} + g_P \rho_P + \frac{\delta E_{xc}}{\delta \rho_P}, \quad \mathcal{V}_A = \mathsf{v}_A + g_V j_A + \frac{\delta E_{xc}}{\delta j^A}, \quad \mathcal{A}_A = \mathsf{a}_A + g_A j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A}.$$

The variation uses $\delta\rho_S/\delta\varphi^\dagger = \beta\varphi$, $\delta\rho_P/\delta\varphi^\dagger = \beta i\chi\varphi$, $\delta j^A/\delta\varphi^\dagger = \Gamma^0\Gamma^A\varphi$, $\delta j_5^A/\delta\varphi^\dagger = \Gamma^0\Gamma^A\chi\varphi$, so the Hartree part is $v_H = \beta g_S \rho_S + \beta i\chi g_P \rho_P + \Gamma^0\Gamma^A g_V j_A + \Gamma^0\Gamma^A \chi g_A j_{5A}$ (checked as Wirtinger derivatives, $\partial e_H / \partial \varphi_a^* = (v_H \varphi)_a$, with 32 independent variables). **Decision (no sea).** The occupied orbitals are the N lowest positive-energy states; the total energy is

$$E = \sum_k \varepsilon_k - \int \big[(\mathcal{S} - \mathsf{s}) \rho_S + (\mathcal{P} - \mathsf{p}) \rho_P + (\mathcal{V}_A - \mathsf{v}_A) j^A + (\mathcal{A}_A - \mathsf{a}_A) j_5^A \big] + E_H + E_{xc}.$$

3.11 Exchange (Wick) and the Fierz identities

For a Slater determinant of N orbitals and contact vertices, Wick's theorem with $\varrho_{xy} = \sum_k \varphi_{k,x} \varphi_{k,y}^*$ and $R = \varrho\Gamma^0$ gives

$$\langle : (\bar\psi O_1 \psi)(\bar\psi O_2 \psi) : \rangle = \operatorname{tr}(O_1 R) \operatorname{tr}(O_2 R) - \operatorname{tr}(O_1 R O_2 R), \qquad R = \sum_k \varphi_k \bar\varphi_k,$$
$$E_x = -\frac{g_S}{2} \operatorname{tr}(RR) - \frac{g_P}{2} \operatorname{tr}(i\chi R \, i\chi R) - \frac{g_V}{2} \operatorname{tr}(\Gamma^A R \Gamma_A R) - \frac{g_A}{2} \operatorname{tr}(\Gamma^A \chi R \Gamma_A \chi R),$$
$$\Sigma_x = -g_S R - g_P (i\chi) R (i\chi) - g_V \Gamma_A R \Gamma^A - g_A \Gamma_A \chi R \Gamma^A \chi, \qquad \delta E_x = \operatorname{tr}(\Sigma_x \delta R).$$

Decision (check). 7+1.nb builds the 560-dimensional Fock space of 3 particles in 16 modes, with $c_a^\dagger c_b$ as explicit sparse matrices and $: B(M_1)B(M_2) := B(M_1)B(M_2) - B(M_1 M_2)$, $B(M) = \sum M_{ab} c_a^\dagger c_b$, $M_i = \Gamma^0 O_i$, and checks the formula for all 18 vertex pairs with 3 random orthonormal orbitals (`SeedRandom[71]`, 40 digits). The Fierz completeness $\sum_K (\Gamma_K)_{ab} (\Gamma_K^{-1})_{cd} = 16 \delta_{ad}\delta_{cb}$ gives the rearrangement

$$\operatorname{tr}(O_1 R O_2 R) = \frac{1}{16} \sum_K \operatorname{tr}(\Gamma_K^{-1} R) \operatorname{tr}(O_1 \Gamma_K O_2 R),$$

checked with random complex integer matrices R, δR (`SeedRandom[1771]`). **Uniform matter.** With $R = (\rho_S 1_{16} + \rho\Gamma_0)/16$, using $\chi R \chi = (\rho_S - \rho\Gamma_0)/16$, $\Gamma^A\Gamma_A = 8$ and $\Gamma^A\Gamma_0\Gamma_A = \Gamma_0 + 7 \cdot (-\Gamma_0) = -6\Gamma_0$:

$$\operatorname{tr}(RR) = \frac{\rho_S^2 + \rho^2}{16}, \qquad \operatorname{tr}(i\chi R \, i\chi R) = -\frac{\rho_S^2 - \rho^2}{16},$$
$$\operatorname{tr}(\Gamma^A R \Gamma_A R) = \frac{8\rho_S^2 - 6\rho^2}{16}, \qquad \operatorname{tr}(\Gamma^A \chi R \Gamma_A \chi R) = -\frac{8\rho_S^2 + 6\rho^2}{16},$$
$$E_x = -\frac{g_S}{32}(\rho_S^2 + \rho^2) + \frac{g_P}{32}(\rho_S^2 - \rho^2) - \frac{g_V}{32}(8\rho_S^2 - 6\rho^2) + \frac{g_A}{32}(8\rho_S^2 + 6\rho^2),$$
$$G_S = g_S + \frac{-g_S + g_P - 8g_V + 8g_A}{16}, \qquad G_V = g_V + \frac{-g_S - g_P + 6g_V + 6g_A}{16},$$

$$\frac{g_S}{2}\rho_S^2+\frac{g_V}{2}\rho^2+E_x=\frac{G_S}{2}\rho_S^2+\frac{G_V}{2}\rho^2$$

(the pseudoscalar and axial Hartree terms vanish for uniform matter, where $\rho_P=j_5^A=0$). For $g_S=-10^5$, $g_V=10^4$, $g_P=g_A=0$: $G_S=-10^5+\frac{10^5-8\cdot10^4}{16}=-98750$ and $G_V=10^4+\frac{10^5+6\cdot10^4}{16}=20000$.

3.12 The chiral Kohn–Sham theory

Decision. Use the variables $\kappa=\sum_k\varphi_{kL}^\dagger\varphi_{kR}$, $j_L^A=\sum_k\varphi_{kL}^\dagger\bar\sigma^A\varphi_{kL}$, $j_R^A=\sum_k\varphi_{kR}^\dagger\sigma^A\varphi_{kR}$ ($\sigma^A=(1_8,\sigma_i)$, $\bar\sigma^A=(1_8,-\sigma_i)$), the complex source $\varpi=\mathsf{s}+i\mathsf{p}$ and the complex mass $\mathcal{M}=m+\mathcal{S}+i\mathcal{P}$. From the chiral blocks of section 3.7:

$$\rho_S=\kappa+\kappa^*,\quad \rho_P=i(\kappa-\kappa^*),\quad j^A=j_L^A+j_R^A,\quad j_5^A=j_R^A-j_L^A,\quad \mathsf{s}\rho_S+\mathsf{p}\rho_P=\varpi\kappa+\varpi^*\kappa^*,$$
$$e_H=\frac{g_S-g_P}{2}(\kappa^2+\kappa^{*2})+(g_S+g_P)\kappa\kappa^*+\frac{g_V+g_A}{2}(j_L^Aj_{LA}+j_R^Aj_{RA})+(g_V-g_A)j_L^Aj_{RA},$$
$$W^\dagger h_{\mathrm{KS}}(\vec p)W=\begin{pmatrix}\mathcal{V}_0^L-\sigma_i(p^i+\mathcal{V}_i^L) & \mathcal{M}1_8\\ \mathcal{M}^*1_8 & \mathcal{V}_0^R+\sigma_i(p^i+\mathcal{V}_i^R)\end{pmatrix},\qquad \mathcal{V}_A^{L,R}=\mathcal{V}_A\mp\mathcal{A}_A,$$
$$\mathcal{M}-m=\varpi+\frac{\partial(e_H+e_{xc})}{\partial\kappa},\qquad \text{Hartree: }\ \mathcal{M}-m-\varpi=g_S(\kappa+\kappa^*)-g_P(\kappa-\kappa^*)=g_S\rho_S+ig_P\rho_P.$$

Notation decision. The complex source is called ϖ, not μ: μ is the chemical potential of section 3.13.
Chiral symmetry. $e^{i\theta\chi}=\cos\theta\,1_{16}-\sin\theta\,\omega$ is real and multiplies κ by $e^{2i\theta}$; E_H changes by $\frac{g_S-g_P}{2}[(e^{4i\theta}-1)\kappa^2+(e^{-4i\theta}-1)\kappa^{*2}]$, so the interaction is chirally symmetric exactly for $g_P=g_S$. The check uses $\cos\theta=\frac35$, $\sin\theta=\frac45$ (rational).

3.13 Uniform matter: formulas, parameters and numerical precision

Static uniform matter at rest, all 8 spin states filled to k_F, no sea, no sources: $\mathcal{S}=m^*-m$, $\mathcal{V}_0$ constant, $\mathcal{P}=\mathcal{A}_A=\mathcal{V}_i=0$. The area of the unit sphere $S^6\subset\mathbb{R}^7$ is $|S^6|=2\pi^{7/2}/\Gamma_{\mathrm E}(\frac72)=\frac{16\pi^3}{15}$ ($\Gamma_{\mathrm E}$ is Euler's Gamma function, $\Gamma_{\mathrm E}(\frac72)=\frac{15}{8}\sqrt\pi$), so

$$8\int\frac{d^7k}{(2\pi)^7}f(|\vec k|)=c_7\int_0^\infty k^6f(k)\,dk,\qquad c_7=\frac{8|S^6|}{(2\pi)^7}=\frac{1}{15\pi^4},$$
$$\rho=c_7\frac{k_F^7}{7}=\frac{k_F^7}{105\pi^4},\qquad \rho_S=c_7\int_0^{k_F}\frac{k^6m^*}{\sqrt{k^2+m^{*2}}}dk,\qquad \varepsilon_{\mathrm{kin}}=c_7\int_0^{k_F}k^6\sqrt{k^2+m^{*2}}\,dk,$$
$$m^*=m+g_S\rho_S(m^*),\qquad \mathcal{V}_0=g_V\rho,\qquad \varepsilon_F=\mathcal{V}_0+\sqrt{k_F^2+m^{*2}},$$
$$\varepsilon=\varepsilon_{\mathrm{kin}}-\frac{(m^*-m)^2}{2g_S}+\frac{g_V}{2}\rho^2.$$

The integrals have the closed forms (obtained with `Integrate` in 7+1.nb, checked by differentiation with sympy), with $\operatorname{artanh}y=\frac12\log\frac{1+y}{1-y}$:

$$\int_0^{k_F}\frac{k^6m^*}{\sqrt{k^2+m^{*2}}}dk=\frac{m^*}{48}\Big[k_F\sqrt{k_F^2+m^{*2}}\big(8k_F^4-10k_F^2m^{*2}+15m^{*4}\big)-15m^{*6}\operatorname{artanh}\frac{k_F}{\sqrt{k_F^2+m^{*2}}}\Big],$$
$$\int_0^{k_F}k^6\sqrt{k^2+m^{*2}}\,dk=\frac{1}{384}\Big[k_F\sqrt{k_F^2+m^{*2}}\big(48k_F^6+8k_F^4m^{*2}-10k_F^2m^{*4}+15m^{*6}\big)-15m^{*8}\operatorname{artanh}\frac{k_F}{\sqrt{k_F^2+m^{*2}}}\Big].$$

Hugenholtz–Van Hove. $\partial\varepsilon/\partial m^*=\rho_S-(m^*-m)/g_S=0$ is the gap equation (with $d\varepsilon_{\mathrm{kin}}/dm^*=\rho_S$), so $d\varepsilon/dk_F=c_7k_F^6(\sqrt{k_F^2+m^{*2}}+g_V\rho)$ and, with $d\rho/dk_F=c_7k_F^6$, the chemical potential is $\mu=d\varepsilon/d\rho=\varepsilon_F$; then $\frac{d}{dk_F}\frac{\varepsilon}{\rho}=\frac{c_7k_F^6}{\rho}\big(\mu-\frac{\varepsilon}{\rho}\big)$ vanishes at the minimum of $E/A=\varepsilon/\rho$, where $E/A=\varepsilon_F$.

Decisions (parameters). Units $m=1$. $g_S=-10^5$, $g_V=10^4$, $g_P=g_A=0$ (couplings of mass dimension -6): with these values the energy per particle has a minimum below m at $k_F=0.7566\ldots$, so the example

shows binding and saturation; the Hartree–Fock comparison is made at $k_F = \frac{3}{4}$. **Decisions (precision).** The gap equation is solved by Brent's method with 50-digit working precision (`wp = 50`, accuracy and precision goals 45), the input k_F padded to 60 digits; the saturation condition $\varepsilon/\rho = \varepsilon_F$ is solved by Brent's method on $[0.7, 0.8]$ with 40-digit working precision and goals 32. The checks require $|E/A - \varepsilon_F| < 10^{-25}$ at saturation and compare μ from a symmetric difference quotient (step $h = 10^{-12}$ in k_F) with ε_F to 10^{-15}. These settings were adopted after development runs in which `FindRoot` issued the message `FindRoot::precw` (input precision below the working precision) and its default goals (half the working precision) left residuals larger than the tolerances of the checks.

3.14 The chiral vacuum with the Dirac sea

Decision. To show spontaneous chiral symmetry breaking, keep the Dirac sea with a momentum cutoff $|\vec{p}| < \Lambda$, take $g_P = g_S = g$, $\varpi = 0$, $\mathcal{V} = \mathcal{A} = 0$ and a uniform complex mass $\mathcal{M}$:

$$h_\chi(\vec{p}) = \begin{pmatrix} -\sigma\cdot\vec{p} & \mathcal{M}1_8 \\ \mathcal{M}^* 1_8 & \sigma\cdot\vec{p} \end{pmatrix}, \qquad h_\chi^2 = (\vec{p}^{\,2} + |\mathcal{M}|^2)1_{16}, \qquad \Pi_- = \frac{1}{2}\Big(1_{16} - \frac{h_\chi}{E}\Big), \quad E = \sqrt{\vec{p}^{\,2} + |\mathcal{M}|^2},$$

$$\kappa_{\vec{p}} = \operatorname{tr}\Pi_-^{RL} = -\frac{4\mathcal{M}^*}{E}, \qquad \frac{|S^6|}{(2\pi)^7} = \frac{1}{120\pi^4}, \qquad \kappa = -4\mathcal{M}^* J(|\mathcal{M}|),$$

$$J(M) = \frac{1}{120\pi^4}\int_0^\Lambda \frac{p^6 dp}{\sqrt{p^2+M^2}}, \qquad K(M) = \frac{1}{120\pi^4}\int_0^\Lambda p^6\sqrt{p^2+M^2}\,dp,$$

$$\mathcal{M} = m + 2g\kappa^* = m - 8gJ(|\mathcal{M}|)\mathcal{M}, \qquad \mathcal{E}(\mathcal{M}) = -8K + 8J\,\mathrm{Re}\big[(\mathcal{M}-m)\mathcal{M}^*\big] + 32g|\mathcal{M}|^2J^2.$$

Here Π_-^{RL} is the lower-left 8×8 block of Π_-, equal to $-\mathcal{M}^*1_8/(2E)$, and $\mathcal{E}$ is the sum of the sea eigenvalues $-8K$ minus the double counting $2\mathrm{Re}[(\mathcal{M}-m)\kappa]$ plus $e_H = 2g|\kappa|^2$. For $m = 0$ a solution $\mathcal{M} \neq 0$ needs $1 + 8gJ(|\mathcal{M}|) = 0$, possible exactly when

$$g < g_c = -\frac{1}{8J(0)} = -\frac{90\pi^4}{\Lambda^6}, \qquad J(0) = \frac{\Lambda^6}{720\pi^4}, \qquad K(0) = \frac{\Lambda^8}{960\pi^4}.$$

Decisions (parameters). $\Lambda = 1$; $g = \frac{6}{5}g_c$ (20% beyond the critical coupling, so that $|\mathcal{M}|$ is clearly non-zero); a small explicit mass $m = \frac{1}{100}$ to show that it selects the real positive $\mathcal{M}$, $\mathcal{E}(\mathcal{M}e^{i\phi}) - \mathcal{E}(\mathcal{M}) = 8Jm|\mathcal{M}|(1-\cos\phi) > 0$; the diagonalization check at $\vec{p} = (\frac{3}{10}, -\frac{1}{5}, \frac{1}{10}, \frac{1}{4}, 0, -\frac{3}{20}, \frac{1}{20})$ with $\mathcal{M} = |\mathcal{M}|e^{7i/10}$ (a generic momentum and phase). The gap equation is solved by Brent's method with 40-digit working precision and goals 32.

3.15 Notation decisions

- The coordinates $x_0, \ldots, x_7$ and the metric η of 7+1.nb are kept. Kohn–Sham potentials are calligraphic $(\mathcal{S}, \mathcal{P}, \mathcal{V}_A, \mathcal{A}_A)$, sources sans-serif $(\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A)$.
- The Dirac matrix $\beta = \Gamma^0$ keeps its name; the rapidity of a boost is β_b (Mathematica `β` in `Sβ`), the spinor boost is $\mathcal{B}(\beta_\mathrm{b})$ (Mathematica `Sβ`) and its Lorentz matrix L (Mathematica `Λboost`), so that $\mathcal{S}$ means only the scalar potential and Λ only the cutoff.
- The complex scalar source is $\varpi = \mathsf{s} + i\mathsf{p}$; μ is only the chemical potential.
- The matrices of the two proofs are Y (no real realization for η') and Z (no real chirality); J and K are only the sea integrals; Euler's Gamma function is Γ_E, to distinguish it from the Dirac matrices Γ_A.
- Mathematica names: `Γ17[A]`, `βD`, `αD[i]`, `ω17`, `χ17`, `EL`, `ER`, `Wch`, `ϕL`, `ϕR`, `σW[i]`, `dirac16complex`, `hKS`, `hChiKS`, and so on; in the notebook text cells the same symbols are written in plain Unicode.

3.16 The signature: the author's explanation of 27 October 2026

Source. On 27 October 2026 the author of the project explained that the xAct list $\{7,1,0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1,-1,\ldots,-1)$ is simply the squares of the octonion basis, and asked that this be made clear in all documentation, examples and provenance records. **What it corrected.** The version of 7+1.nb pushed in the commits 52d8729 to dc2f178 of github.com/once-ere/7plus1 declared the xAct metric of section 12 with `DefMetric[{1, 7, 0}, …]`, reading the list as the signs of η; since then section 12 declares `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`, exactly as the author's notebook Einstein-Lovelock-7+1-Nash.nb does. **What was added.** Checks in 7+1.nb — section 4.7: η is the squares of the octonion basis, $e_A e_A = \eta_{AA} e_0$ and $\mathrm{Re}(e_A e_B) = \eta_{AB}$; section 12.2: `SignatureOfMetric[g71]` $= \{7,1,0\}$ with `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$, and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$; section 17: the flat metric $\mathrm{diag}(+1,-1,-1,-1)$ of the world brane is the squares of the basis e_0, e_1, e_2, e_3 of the quaternion subalgebra — and the same statements in the Python programs `tools/ref71.py`, `tools/ref_gravity.py` and `tools/ref_sm.py`; the explanation itself in the text of 7+1.nb (sections 1.2, 1.4, 4.2, 4.7, 7, 11, 12.2, 13, 15.1, 17) and in every document, symbol file and provenance file of the repository. **What did not change.** No result. Compared output by output with the backup below, every output of 7+1.nb is the same, except the declared list itself, the new checks and the check counts (and, as in every run, the name of the private context and the measured times); in particular every component of every Lovelock tensor is unchanged. The components follow from the component matrix of the metric, and the sign of the determinant, `SignDetOfMetric`, is -1 for both lists. **Backup.** The evaluated notebook as it was before is kept in the workspace as `backups/7+1_2026-10-27_before_signature.nb` (the folder backups/ is excluded from the repository by .gitignore), 1034401 bytes, SHA-256 `060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`; `tools/verify_7p1_nb.py` compares every input, text and section cell of 7+1.nb with it and finds every cell present, each one unchanged or changed by this explanation, and it compares every output in the same way.

The explanation, in full. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta' = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta' = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta' = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

4. Verification record

4.1 The Mathematica run

item	value
program	`C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`
$Version	15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)

item	value
method	`tools/run_eval.wl`: all cells evaluated in a fresh kernel (front-end kernel configuration "Eval71"; time limit 3600 s), then `NotebookSave`; requested through `tools/watch_requests.wl` (job `eval`)
finished	Tue 27 Oct 2026 19:31:35
evaluation time of the whole notebook	271.569 s
private context of the notebook	`Notebook$$59$143224`
input cells / output cells / message cells	166 / 453 / 0
checks in sections 7–9 / True	74 / 74
checks in the whole notebook / True	399 / 399
summary of section 10	`Length[checks]` = 203, `failedChecks` = {}, `allChecksPassed` = `True`
SHA-256 of the evaluated 7+1.nb (1849785 bytes)	`1314b1e09e6b6eefbd9ee5974e19d186a5e996907b2eb68ddb6f23aae85aaaa5`

Before the final evaluation, every input cell of sections 7–9 was developed with `tools/eval_cells_log.wl`, which evaluates a generated notebook expression by expression in a fresh context, records the time, the messages and the result of every check, and stops at the first failure.

4.2 The checks of sections 7–9 of 7+1.nb and their results

Each row is one cell `check["name", test]` of the notebook, in order; the test is the Wolfram Language expression in that cell, and the result is what Wolfram 15.0.1 returned.

7.1 The eight real Dirac matrices and the Dirac α, β

#	check	result
1	`Dirac matrices: Γ17up[0] == Γ17[0], Γ17up[i] == -Γ17[i]; all REAL, every entry 0 or ±1`	True
2	`Dirac adjoint: ConjugateTranspose[Γ17up[A]] == Γ17[0].Γ17up[A].Γ17[0] for A = 0..7`	True
3	`βD == {{0,ID8},{ID8,0}} and αD[i] == {{0,τO[i]},{-τO[i],0}}: real and symmetric`	True
4	`αD[i].αD[j] + αD[j].αD[i] == 2 δij ID16, αD[i].βD + βD.αD[i] == 0, βD.βD == ID16`	True
5	`ω17 = Γ17[8] == {{0,-ID8},{ID8,0}} is real, ω17.ω17 == -ID16, and ω17 anticommutes with every Γ17[A]`	True

7.2 The field dirac16complex and its Dirac equation

#	check	result
6	`(I Γ17up[A] ∂_A + m)(I Γ17up[A] ∂_A - m) dirac16complex == -(∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) dirac16complex (16 arbitrary functions)`	True

7.3 Hamiltonian form

#	check	result
7	`Γ17[0].(I Γ17up[A] ∂_A - m) ψ == I ∂0 ψ - h ψ with h = -I αD[i] ∂_i + βD m (16 arbitrary functions)`	True

7.4 Plane waves and the real spinors u_s, v_s

#	check	result
8	`h(p) == {{0, M}, {Transpose[M], 0}} with the REAL 8×8 matrix M = m ID8 + p_i τO[i]`	True

#	check	result
9	M.Transpose[M] == Transpose[M].M == E^2 ID8 and h(p).h(p) == E^2 ID16, E^2 = m^2 + p^2	True
10	h(p).u_s == E u_s and h(-p).v_s == -E v_s for s = 1..8	True
11	(pslash - m).u_s == 0 and (pslash + m).v_s == 0, pslash = E Γ17[0] + p_i Γ17[i]	True
12	u_s.u_r == v_s.v_r == 2E δsr, and u_s(p).v_r(-p) == 0	True
13	plane waves: u_s Exp[-I(E x0 - p.x)] and v_s Exp[+I(E x0 - p.x)] solve (I Γ17up[A] ∂_A - m) ψ == 0	True
14	example p = (2,2,4,0,0,0,0), m = 1: E = 5 and the eight u_s are real (entries in Sqrt[5] ℚ) and solve h(p).u_s == 5 u_s	True

7.5 Spin sums, completeness, and the diagonalization of h

#	check	result
15	spin sums: Σ_s u_s ubar_s == pslash + m and Σ_s v_s vbar_s == pslash - m	True
16	completeness: Σ_s u_s(p) u_s(p)^T + Σ_s v_s(-p) v_s(-p)^T == 2E ID16	True
17	U = (u_1..u_8, v_1(-p)..v_8(-p))/Sqrt[2E] is orthogonal and Transpose[U].h(p).U == diag(E ID8, -E ID8)	True

7.6 Grassmann algebra: anticommuting components, Lagrangian, conserved current

#	check	result
18	Grassmann: θ1 θ2 == -θ2 θ1, θ1 θ1 == 0, associativity, and ∂/∂θ2 (θ1 θ2 θ3) == -θ1 θ3	True
19	the three plane waves U1, U2, U3 solve the Dirac equation with m = 1	True
20	Grassmann field: ψ_a ψ_b + ψ_b ψ_a == 0 for all a, b (in particular ψ_a^2 == 0)	True
21	current conservation ∂_A (ψbar Γ17up[A] ψ) == 0 for the Grassmann-valued solution ψ = θ1 U1 + θ2 U2 + θ3 U3	True

7.7 Quantization: mode expansion and canonical anticommutators

#	check	result
22	canonical anticommutator from the mode algebra: {ψ_a(x0,x), ψ_b(x0,y)^†} == δab (1/V) Σ_p Exp[I p.(x - y)] (five momenta, symbolic x, y)	True

7.8 Charge conjugation

#	check	result
23	ψ -> ψc = ω17.ψ^*: (I Γ17up[A] ∂_A - m)(ω17.ψ^*) == ω17.((I Γ17up[A] ∂_A - m) ψ)^* for arbitrary ψ = ψre + I ψim	True
24	(pslash + m).ω17.u_s == 0 and ω17.u_s == Σ_r v_r (-Transpose[M]/E)_rs; ω17.ω17 == -ID16 (no Majorana spinors)	True

7.9 Lorentz covariance

#	check	result
25	boost at β = 2 log 2: S == 5/4 ID16 + 3/4 Γ17[0].Γ17[1], p' = Λ.p == (55/8, -41/8, 2, 4, 0, 0, 0, 0) is on shell, (pslash' - m).S.u_s(p) == 0	True

7.10 Chirality: no real chirality matrix, complex chiral projections

#	check	result
26	χ17 = I ω17: Hermitian, χ17.χ17 == ID16, anticommutes with every Γ17[A], commutes with every Σ17[A,B]	True
27	PL17, PR17 are complex (not real); Γ17[A].PL17 == PR17.Γ17[A]; ConjugateTranspose[PL17].Γ17[0] == Γ17[0].PR17	True
28	EL, ER: χ17.EL == -EL, χ17.ER == ER, EL^†.EL == ER^†.ER == ID8, EL^†.ER == 0, EL.EL^† == PL17, ER.ER^† == PR17, W unitary	True
29	φL == (ψup + I ψdown)/Sqrt[2], φR == (I ψup + ψdown)/Sqrt[2], PL17.ψ == EL.φL, PR17.ψ == ER.φR, ψ == EL.φL + ER.φR	True

7.11 The two coupled 8-component equations

#	check	result
30	σW[i] = I τO[i]: Hermitian, σW[i].σW[j] + σW[j].σW[i] == 2 δij ID8	True
31	chiral (Weyl) form: W^†.h(p).W == {{-σW.p, m ID8}, {m ID8, σW.p}} (symbolic p)	True
32	EL^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 + τO[i] ∂_i) φL - m φR (16 arbitrary functions)	True
33	ER^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 - τO[i] ∂_i) φR - m φL (16 arbitrary functions)	True
34	Klein-Gordon for each 8-component field: (I ∂0 - τO[i] ∂_i)(I ∂0 + τO[j] ∂_j) f == -(∂0^2 - ∂1^2 - ... - ∂7^2) f (8 arbitrary functions)	True
35	plane-wave example p = (2,2,4,0,0,0,0), m = 1, E = 5: (E + σW.p) φL == m φR and (E - σW.p) φR == m φL for all eight u_s	True

8.1 The self-coupled field: Lagrangian, densities, Hamiltonian

#	check	result
36	the densities ρS, ρP, j^A, j5^A are real: Γ17[0].op is Hermitian for op = ID16, I χ17, Γ17up[A], Γ17up[A].χ17	True
37	18 vertex pairs; lower vertex == η17[[A+1,A+1]] upper vertex in the V and A channels	True

8.2 The nonlinear field equation (Grassmann left derivatives)

#	check	result
38	nonlinear field equation: left derivative ∂L_int/∂ψbar_a == -[gS (ψbar ψ) ψ + gP (ψbar Iχ17 ψ) Iχ17 ψ + gV (ψbar Γ17[A] ψ) Γ17up[A] ψ + gA (ψbar Γ17[A]χ17 ψ) Γ17up[A]χ17 ψ]_a (32 Grassmann generators)	True

8.4 The 16-component Kohn–Sham equations

#	check	result
39	h_KS(p) is Hermitian for real S, P, V_A, A_A (symbolic momentum and potentials)	True
40	plane wave: Γ17[0].(pslash(ε,p) - m - S - I χ17 P - Γ17up[A](V_A + χ17 A_A)) == ε ID16 - h_KS(p)	True
41	Hartree potentials: ∂E_H/∂φ_a^* == (v_H.φ)_a, v_H = h_KS - h with S = gS ρS, P = gP ρP, V_A = gV j_A, A_A = gA j5_A (32 variables)	True

8.5 Exchange: Wick's theorem for contact couplings

#	check	result
42	Wick's theorem in the 560-dimensional Fock space: <Φ| :(ψbar o1 ψ)(ψbar o2 ψ): |Φ> == tr(o1 R) tr(o2 R) - tr(o1 R o2 R) for all 18 vertex pairs	True

8.6 The exchange self-energy and the Fierz identities

#	check	result
43	exchange self-energy: d/dt E_x(R + t dR) at t = 0 == tr(Σ_x.dR) and E_x == tr(Σ_x.R)/2 (random complex R, dR; symbolic couplings)	True
44	the 256 products Γ_K are orthogonal matrices with tr(Inverse[Γ_K].Γ_L) == 16 δKL	True
45	Fierz completeness: Σ_K (Γ_K)_ab (Inverse[Γ_K])_cd == 16 δad δcb (as the 256×256 matrix identity Transpose[F].F == 16 ID)	True
46	Fierz rearrangement of every exchange term: tr(o1 R o2 R) == (1/16) Σ_K tr(Inverse[Γ_K].R) tr(o1.Γ_K.o2.R) for all 18 vertex pairs (random complex R)	True

8.7 Example: uniform matter (no sea), Hartree and Hartree–Fock

#	check	result
47	uniform matter: tr R == ρS, tr(Γ17up[0].R) == ρ, and the densities ρP, j^i, j5^A vanish	True
48	the occupied plane waves give this R: (1/2) Σ_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) == (E Γ17[0] + m ID16)/(2E)	True
49	uniform exchange energy: E_x(R) == -gS/32 (ρS^2 + ρ^2) + gP/32 (ρS^2 - ρ^2) - gV/32 (8ρS^2 - 6ρ^2) + gA/32 (8ρS^2 + 6ρ^2)	True
50	uniform exchange self-energy: Σ_x == S_x ID16 + V_x Γ17[0] with S_x == ∂E_x/∂ρS and V_x == ∂E_x/∂ρ	True
51	uniform Hartree-Fock == Hartree with the Fierz-transformed couplings: gS/2 ρS^2 + gV/2 ρ^2 + E_x == G_S/2 ρS^2 + G_V/2 ρ^2 (gP, gA arbitrary)	True
52	c7 == 8 |S^6|/(2 Pi)^7 == 1/(15 Pi^4), |S^6| == 2 Pi^(7/2)/Gamma[7/2] == 16 Pi^3/15, and ρ == c7 kF^7/7 == kF^7/(105 Pi^4)	True
53	closed forms: d/dkF of the integrals gives the integrands, both vanish at kF = 0, and dε_kin/dm* == ρS	True
54	Hartree gap equation m* == m + gS ρS(m*) solved to 45 digits; m* is a stationary point of ε(m*) at fixed kF	True
55	total energy == sum of KS eigenvalues - E_H (double counting) == ε_kin - (m* - m) ρS + E_H	True
56	Hugenholtz-Van Hove at saturation: E/A == ε_F == dε/dρ; E/A - m < 0 and E/A at 0.95 kF0 and 1.05 kF0 is higher	True
57	Hugenholtz-Van Hove away from saturation: μ == dε/dρ == ε_F at kF = 1	True
58	Hartree-Fock at kF = 3/4 (gP = gA = 0): G_S == -98750, G_V == 20000; m* - m == gS ρS + S_x, V_0 == gV ρ + V_x; μ == ε_F	True

9.1 Chiral densities

#	check	result
59	chiral densities: ρS == κ + κ^*, ρP == I(κ - κ^*), j^A == jL^A + jR^A, j5^A == jR^A - jL^A (symbolic chiral orbitals)	True
60	the chiral blocks behind these densities: EL^†.βD.ER == ID8, EL^†.βD.EL == 0, EL^†.αD[i].EL == -σW[i], ER^†.αD[i].ER == σW[i], EL^†.αD[i].ER == 0	True

9.2 The Hartree energy in chiral variables

#	check	result
61	E_H (16-component form, chiral orbitals) == gS/2 (κ + κ^*)^2 - gP/2 (κ - κ^*)^2 + (gV + gA)/2 (jL.jL + jR.jR) + (gV - gA) jL.jR	True

9.3 The chiral Kohn–Sham equations

#	check	result
62	chiral KS Hamiltonian: W^†.h_KS(p).W == {{V^L_0 - σW.(p + V^L), M}, {M^*, V^R_0 + σW.(p + V^R)}}, M = m + S + I P, V^L = V - A, V^R = V + A	True
63	Hartree (Wirtinger): M - m == ∂e_H/∂κ == gS ρS + I gP ρP, V^L_A == ∂e_H/∂jL^A; V^L_A == V_A - A_A and V^R_A == V_A + A_A with V_A = gV j_A, A_A = gA j5_A	True
64	chiral KS potentials are derivatives: ∂E_H/∂φL^* == (V^L_0 - σW[i] V^L_i) φL + (M - m) φR and ∂E_H/∂φR^* == (V^R_0 + σW[i] V^R_i) φR + (M - m)^* φL	True

9.4 Chiral symmetry

#	check	result
65	e^(Iθχ17) == cosθ ID16 - sinθ ω17 is REAL and orthogonal; U.EL == e^(-Iθ) EL, U.ER == e^(Iθ) ER	True
66	U^†.βD.U == βD.(cos 2θ ID16 + I sin 2θ χ17); vector and axial densities invariant	True
67	E_H(e^(Iθχ17) ψ) - E_H(ψ) == (gS - gP)/2 [(e^(4Iθ) - 1) κ^2 + (e^(-4Iθ) - 1) κ^*2]: U(1)_A symmetric iff gP == gS	True

9.5 Uniform matter in chiral form

#	check	result
68	uniform matter in chiral form: κ == ρS/2, jL^0 == jR^0 == ρ/2, jL^i == jR^i == 0	True
69	(h_χ - V_0)^2 == (p^2 + m*^2) ID16: eigenvalues V_0 ± Sqrt[p^2 + m*^2], eight times each	True

9.6 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking

#	check	result
70	|S^6|/(2 Pi)^7 == 1/(120 Pi^4); J(0) == Λ^6/(720 Pi^4); g_c == -1/(8 J(0)) == -90 Pi^4/Λ^6; J(M) -> J(0) and K(M) -> K(0) as M -> 0	True
71	m = 0, g = 6/5 g_c: the gap equation 1 + 8 g J(|M|) == 0 has the root |M| (32 digits); M == 2 g κ^* with κ == -4 M J; E is stationary there and lower than E(0)	True
72	degenerate chiral vacua for m = 0: E(|M| e^(I ϕ)) == E(|M|) for ϕ = 1, 2, 4 (the phase is the Goldstone direction)	True
73	explicit breaking m = 1/100: the real M == m - 8 g J(M) M solves the gap equation, E is stationary there, and every complex phase raises E	True
74	the sea at one momentum: h_χ^2 == E^2 ID16; Π_- is a projector of rank 8; κ_p == tr(Π_- RL block) == -4 M^*/E; tr(Π_-.h0) == -8E + 8 Re[(M - m) M^*]/E	True

4.3 Independent Python checks: tools/ref_dirac16.py

The program recomputes the constructions of sections 7–9 without Mathematica (Python 3.13.17, sympy 1.14.0, mpmath 1.3.0, numpy 2.5.3; its random test data come from the seed 71 of `random` and of numpy's `default_rng`). Its 90 boolean checks:

#	check	result
1	Gamma17 real, entries 0,+-1	True
2	{Gamma_A,Gamma_B} = 2 eta17_AB	True
3	Gamma0 symmetric, Gamma_i antisymmetric	True
4	Gamma^A dagger = Gamma0 Gamma^A Gamma0	True
5	alpha^i, beta real symmetric	True
6	{alpha_i,alpha_j}=2delta, {alpha_i,beta}=0, beta^2=1	True
7	block forms alpha^i = [[0,tauO_i],[-tauO_i,0]], beta = [[0,1],[1,0]]	True
8	omega = Gamma17[8] = [[0,-1],[1,0]], omega^2 = -1	True
9	(i Dslash + m)(i Dslash - m) psi = -(box + m^2) psi	True
10	Gamma0 (i d0 - h) psi = (i Dslash - m) psi	True
11	h(p) = [[0,M],[M^T,0]]	True
12	h(p)^2 = E^2 1	True
13	M M^T = M^T M = E^2 1_8	True
14	h(p) u_s = E u_s, h(-p) v_s = -E v_s	True
15	(pslash - m) u_s = 0, (pslash + m) v_s = 0	True
16	u_s^T u_r = 2E delta, v_s^T v_r = 2E delta	True
17	u_s(p)^T v_r(p~) = 0	True
18	sum_s u_s ubar_s = pslash + m, sum_s v_s vbar_s = pslash - m	True
19	completeness sum u u^T + sum v(p~) v(p~)^T = 2E 1_16	True
20	U_p orthogonal and U_p^T h(p) U_p = diag(E 1_8, -E 1_8)	True
21	charge conjugation: (pslash + m) omega u_s = 0	True
22	omega u_s = sum_r v_r (-M^T/E)_rs	True
23	S Gamma_K S^-1 = sum_B Lambda_BK Gamma_B (boost)	True
24	Lorentz covariance: (pslash(Lambda p) - m) S u_s(p) = 0	True
25	real matrices anticommuting with all Gamma17[A]: dimension 1 (rank 255)	True
26	omega anticommutes with all Gamma17[A]	True
27	chi = i omega: Hermitian, chi^2 = 1, {chi,Gamma_A} = 0	True
28	[chi, Sigma_AB] = 0 (chirality is Lorentz invariant)	True
29	P_L, P_R not real	True
30	P_L+P_R = 1, P_L^2 = P_L, P_R^2 = P_R, P_L P_R = 0, rank 8 each	True
31	Gamma_A P_L = P_R Gamma_A, P_L^dagger Gamma0 = Gamma0 P_R	True
32	E_L, E_R isometries: E^dag E = 1_8, E_L^dag E_R = 0, E_L E_L^dag = P_L, E_R E_R^dag = P_R	True
33	chi E_L = -E_L, chi E_R = E_R, W unitary	True
34	W^dag h(p) W = [[-sigma.p, m],[m, sigma.p]]	True
35	sigma_i Hermitian, {sigma_i,sigma_j} = 2 delta	True
36	chiral blocks: E_L^dag beta E_L = 0, E_L^dag beta E_R = 1, E_L^dag alpha E_L = -sigma, E_R^dag alpha E_R = sigma, E_L^dag alpha E_R = 0	True
37	E_L^dag Gamma0 (i Dslash - m) psi = (i d0 + tauO.d) phi_L - m phi_R	True
38	E_R^dag Gamma0 (i Dslash - m) psi = (i d0 - tauO.d) phi_R - m phi_L	True
39	(i d0 - tauO.d)(i d0 + tauO.d) f = -box f (8 components)	True
40	psi = E_L phi_L + E_R phi_R	True
41	chiral form of the bilinears (beta i chi, chi, alpha chi)	True
42	{psi_a(t,x), psi_b^dag(t,y)} = delta_ab (1/V) sum_q e^{iq(x-y)} (5 momenta, abstract mode algebra)	True
43	{psi_a(t,x), psi_b(t,y)} = 0	True
44	the three U_k solve the Dirac equation (m = 1)	True
45	Grassmann components anticommute: psi_a psi_b + psi_b psi_a = 0, psi_a^2 = 0	True
46	current conservation d_A (psibar Gamma^A psi) = 0 (Grassmann field)	True
47	dL_int/dpsibar_a = -[gS(psibar psi) psi + gP (psibar i chi psi) i chi psi + gV (psibar G_A psi) G^A psi + gA (psibar G_A chi psi) G^A chi psi]_a	True
48	h_KS Hermitian for real S, P, V_A, A_A	True
49	Gamma0 [pslash(eps,p) - m - S - i chi P - Gamma^A (V_A + chi A_A)] = eps - h_KS(p) (plane wave e^{-i(eps t - p.x)})	True
50	W^dag h_KS W = [[V^L_0 - sigma.(p+V^L), M],[M^*, V^R_0 + sigma.(p+V^R)]], V^L = V - A, V^R = V + A, M = m + S + iP	True
51	Hartree potentials: dE_H/dphi*_a = (v_KS phi)_a with S = gS rhoS, P = gP rhoP, V_A = gV j_A, A_A = gA j5_A	True
52	densities real: Gamma0 O Hermitian for O = 1, i chi, Gamma^A, Gamma^A chi	True

#	check	result
53	chiral densities: rhoS = kappa + kappa^*, rhoP = i(kappa - kappa^*), j = jL + jR, j5 = jR - jL	True
54	E_H = gS/2 (k+k*)^2 - gP/2 (k-k*)^2 + (gV+gA)/2 (jL.jL + jR.jR) + (gV-gA) jL.jR	True
55	V^L_A = V_A - A_A, V^R_A = V_A + A_A (Hartree, chiral currents)	True
56	M - m = dE_H/dkappa = gS(k+k*) - gP(k-k*) = S + iP	True
57	dE_H/dphi_L* = (V^L_0 - sigma_i V^L_i) phi_L + (M - m) phi_R ; dE_H/dphi_R* = (V^R_0 + sigma_i V^R_i) phi_R + (M - m)^* phi_L	True
58	U_theta = cos(theta) - sin(theta) omega is real orthogonal; U E_L = e^{-i theta} E_L, U E_R = e^{i theta} E_R	True
59	U^dag beta U = beta (cos 2theta + i sin 2theta chi); vector and axial currents invariant	True
60	E_H(e^{i theta chi} psi) - E_H(psi) = (gS-gP)/2 [(e^{4i theta}-1) k^2 + (e^{-4i theta}-1) k*^2] -> U(1)_A symmetric iff gP = gS	True
61	256 products Gamma_K are orthogonal: Gamma_K^-1 = Gamma_K^T	True
62	tr(Gamma_K^-1 Gamma_L) = 16 delta_KL (256 x 256)	True
63	Fierz completeness: sum_K (Gamma_K)_ab (Gamma_K^-1)_cd = 16 delta_ad delta_cb	True
64	delta E_x = tr[Sigma_x delta R] (random R, dR), Sigma_x = -gS R - gP (i chi) R (i chi) - gV G_A R G^A - gA G_A chi R G^A chi	True
65	E_x = (1/2) tr[Sigma_x R]	True
66	E_H = (gS/2)(tr R)^2 + ... = sum_c (g_c/2) tr(O_c R) tr(O^c R) and dE_H = tr[Sigma_H dR]	True
67	Fierz: tr[O1 R O2 R] = (1/16) sum_K tr(Gamma_K^-1 R) tr(O1 Gamma_K O2 R) for all channel vertices	True
68	uniform R: tr R = rho_S, tr(Gamma^0 R) = rho	True
69	(1/2) sum_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) = (E Gamma_0 + m)/(2E)	True
70	E_x(uniform) = -gS/32 (rS^2+rho^2) + gP/32 (rS^2-rho^2) - gV/32 (8rS^2-6rho^2) + gA/32 (8rS^2+6rho^2)	True
71	Sigma_x(uniform) = S_x + V_x Gamma_0, S_x = rS/16(-gS+gP-8gV+8gA), V_x = rho/16(-gS-gP+6gV+6gA)	True
72	S_x = dE_x/drho_S, V_x = dE_x/drho	True
73	uniform Hartree-Fock = Hartree with G_S = gS + (-gS+gP-8gV+8gA)/16, G_V = gV + (-gS-gP+6gV+6gA)/16	True
74	uniform matter: Hartree terms of P and A vanish (rho_P = 0, j5 = 0, j^i = 0)	True
75	Wick (3 particles, 16 modes): <:(psibar O psi)(psibar O' psi):> = tr(O R) tr(O' R) - tr(O R O' R) for all 18 vertex pairs	True
76	c7 = 8 |S^6|/(2pi)^7 = 1/(15 pi^4) and rho = c7 kF^7/7 = kF^7/(105 pi^4)	True
77	Hartree gap: m* - m - gS rhoS(m*) = 0	True
78	m* is a stationary point of eps(m*) = ekin(m*) - (m*-m)^2/(2gS) + gV rho^2/2	True
79	total energy = sum eps_k - E_H (double counting) = ekin - (m*-m) rhoS + E_H	True
80	Hugenholtz-Van Hove at saturation: E/A = eps_F = d eps/d rho	True
81	d(eps/rho)/dkF = 0 at kF0 (minimum: E/A(0.95 kF0), E/A(1.05 kF0) larger)	True
82	Hugenholtz-Van Hove mu = d eps/d rho = eps_F at kF = 1 (not saturated)	True
83	HF self-consistency with the exchange self-energy: m* - m = gS rhoS + S_x, V0 = gV rho + V_x	True
84	HF energy: ekin - (m*-m) rhoS + E_H + E_x = eps(G_S, G_V)	True
85	HF Hugenholtz-Van Hove: d eps/d rho = eps_F at kF = 3/4	True
86	J(0) = Lambda^6/(720 pi^4), g_c = -1/(8 J(0)) = -90 pi^4/Lambda^6	True
87	NJL m = 0: gap 1 + 8 g J(|M|) = 0 solved, M = 2 g kappa^*, stationary and lower than M = 0	True
88	NJL m = 0: E(|M| e^{i theta}) independent of theta (degenerate chiral vacua)	True
89	NJL m = 1/100: M = m - 8 g J(M) M real, stationary; E(M e^{i theta}) > E(M) for theta != 0	True
90	sea at one momentum: eigenvalues -E (x8), +E (x8); kappa_p = tr[(Pi_-)_{RL}] = -4 M^*/E; tr[Pi_- h_0] = -8E + 8 Re[(M-m) M^*]/E	True

The numbers it computes (30-digit arithmetic). `tools/verify_7p1_nb.py`, whose output is printed below together with that of `tools/ref_dirac16.py`, compares them with the numbers printed by 7+1.nb:

quantity	value
Hartree saturation kF0	0.75663615070149289469
Hartree rho0	0.00001388100213231687719 7
Hartree m*	0.34809734014624326514
Hartree rhoS	$6.5190265985375673486e-6$
Hartree V0	0.13881002132316877197
Hartree E/A - m	−0.028321427964738575854
Hartree eps_F	0.97167857203526142415
Hartree kF=1: m*	0.080964430277606126329

quantity	value
Hartree kF=1: E/A - m	0.41086484863433217115
Hartree kF=1: eps_F	1.9809848613081542876
HF couplings G_S, G_V	$(mpf('-98750.0'), mpf('20000.0'))$
HF kF=3/4: m*	0.36643630198154392124
HF kF=3/4: V0	0.26101775473161552570
HF kF=3/4: E/A - m	0.039146338016119861526
Hartree kF=3/4: m*	0.36297394764803019822
Hartree kF=3/4: E/A - m	−0.028065375846872960879
NJL g_c	−8766.8181930602193513
NJL g = 6/5 g_c	−10520.181831672263222
NJL |M| (m = 0)	0.55369385182924169122
NJL kappa = -4 M J(|M|) (M real)	−0.000026315792858364874200
NJL E(|M|) - E(0)	$-1.3102449629010022473e-6$
NJL M (m = 1/100)	0.58531284674622051123

tools/verify_7p1_nb.py (parses the evaluated 7+1.nb; exit code 0)

```
input cells
    ...: 166
output cells
    ...: 453
message cells (must be 0)
    ...: 0
check outputs
    ...: 363
check outputs that are True
    ...: 363
check["..."] calls in the input cells
    ...: 363
$Version output
    ...: ['{"15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)","Notebook$$59$143224`"}']
private notebook context
    ...: True
Length[checks]
    ...: 399
failedChecks == {}
    ...: True
allChecksPassed output True
    ...: True
tau[A] printed == 4+4 tau[A] (python)
    ...: True
tauO[A] printed == python tauO[A]
    ...: True
Gamma17[A] printed == python G17[A]
    ...: True
gridO printed == python octonion table
    ...: True
gridx44 printed == python split table (4+4.nb)
    ...: True
Lambda boost/rotation printed
    ...: True
8 real on-shell spinors printed solve (pslash - m) w = 0 (python check)
    ...: True
classification table printed
    ...: True
section 7: check outputs (all True)
    ...: 35 (True)
section 8: check outputs (all True)
    ...: 23 (True)
section 9: check outputs (all True)
    ...: 16 (True)
section 7: printed Sqrt[5] u_s(p) == python (5 e_s; M^T e_s) and (pslash - m) u_s == 0
```

```
    ...: True
section 8: printed closed forms of int k^6 m*/E and int k^6 E: d/dkF == integrands, zero at kF
    ...= 0 (sympy)
    ...: True
sections 8-9: 23 printed numbers agree with tools/ref_dirac16.py to 17 digits
    ...: True
sections 8-9: G_S == -98750, G_V == 20000 printed
    ...: True
tools/ref_dirac16.py: all its boolean checks True
    ...: True
sections 0-6: the 166 input and text cells of the backup are all present, 159 unchanged, 7
    ...changed and 1 added by the signature explanation of 27 October 2026
    ...: True
section 11: check outputs (all True)
    ...: 7 (True)
section 12: check outputs (all True)
    ...: 28 (True)
section 13: check outputs (all True)
    ...: 15 (True)
section 14: check outputs (all True)
    ...: 14 (True)
section 15: check outputs (all True)
    ...: 16 (True)
section 16: check outputs (all True)
    ...: 8 (True)
section 17: check outputs (all True)
    ...: 11 (True)
section 18: check outputs (all True)
    ...: 7 (True)
section 19: check outputs (all True)
    ...: 6 (True)
summary table: entries == Length[checks], all True
    ...: True
summary table: every printed check is listed
    ...: True
sections 0-10: the 368 input and text cells of the backup taken before sections 11-20 are all
    ...present, 360 unchanged, 8 changed and 1 added by the signature explanation of 27 October
    ...2026
    ...: True
sections 0-20: the 601 input and text cells of the backup taken before the signature
    ...explanation are all present, 586 unchanged, 15 changed and 4 added by that explanation
    ...: True
sections 0-20, outputs: of the 382 outputs of the backup taken before the signature explanation
    ...373 are identical; the others differ only in the check counts and tables of the summary
    ...sections, the declared list {7,1,0}, the private context name and measured times; 5 new
    ...check outputs, all True : True
sections 14-19: 27 groups of printed numbers agree with the Python programs (relative 1e-6;
    ...rounded prints 1e-4; ODE and ill-conditioned results 1e-5)
    ...: True
section 20: check outputs (all True)
    ...: 14 (True)
section 21: check outputs (all True)
    ...: 19 (True)
section 22: check outputs (all True)
    ...: 9 (True)
section 23: check outputs (all True)
    ...: 6 (True)
stage 4: without sections 20-23 (and with the old title of the summary section) the 605 input
    ...and text cells equal those of the backup taken before sections 20-23 were appended
    ...: True
stage 4: the 387 outputs of sections 0-19 and of the summary equal those of the backup (summary
    ...counts and tables, the private context name and measured times excepted)
    ...: True
sections 20-23: 34 groups of printed numbers agree with tools/ref_coupled.py (tolerances per
    ...group, from 1e-14 for exact constants to 2e-3 for numbers read off numerical solutions)
```

```
    ...: True

VERDICT: ALL CROSS-CHECKS PASSED
```

tools/ref_dirac16.py (independent checks of sections 7-9; exit code 0)

```
Gamma17 real, entries 0,+-1 : True
{Gamma_A,Gamma_B} = 2 eta17_AB : True
Gamma0 symmetric, Gamma_i antisymmetric : True
Gamma^A dagger = Gamma0 Gamma^A Gamma0 : True
alpha^i, beta real symmetric : True
{alpha_i,alpha_j}=2delta, {alpha_i,beta}=0, beta^2=1 : True
block forms alpha^i = [[0,tauO_i],[-tauO_i,0]], beta = [[0,1],[1,0]] : True
omega = Gamma17[8] = [[0,-1],[1,0]], omega^2 = -1 : True
(i Dslash + m)(i Dslash - m) psi = -(box + m^2) psi : True
Gamma0 (i d0 - h) psi = (i Dslash - m) psi : True
h(p) = [[0,M],[M^T,0]] : True
h(p)^2 = E^2 1 : True
M M^T = M^T M = E^2 1_8 : True
h(p) u_s = E u_s,  h(-p) v_s = -E v_s : True
(pslash - m) u_s = 0, (pslash + m) v_s = 0 : True
u_s^T u_r = 2E delta, v_s^T v_r = 2E delta : True
u_s(p)^T v_r(p~) = 0 : True
sum_s u_s ubar_s = pslash + m, sum_s v_s vbar_s = pslash - m : True
completeness sum u u^T + sum v(p~) v(p~)^T = 2E 1_16 : True
U_p orthogonal and U_p^T h(p) U_p = diag(E 1_8, -E 1_8) : True
charge conjugation: (pslash + m) omega u_s = 0 : True
omega u_s = sum_r v_r (-M^T/E)_rs : True
S Gamma_K S^-1 = sum_B Lambda_BK Gamma_B (boost) : True
Lorentz covariance: (pslash(Lambda p) - m) S u_s(p) = 0 : True
real matrices anticommuting with all Gamma17[A]: dimension 1 (rank 255) : True
omega anticommutes with all Gamma17[A] : True
chi = i omega: Hermitian, chi^2 = 1, {chi,Gamma_A} = 0 : True
[chi, Sigma_AB] = 0 (chirality is Lorentz invariant) : True
P_L, P_R not real : True
P_L+P_R = 1, P_L^2 = P_L, P_R^2 = P_R, P_L P_R = 0, rank 8 each : True
Gamma_A P_L = P_R Gamma_A,  P_L^dagger Gamma0 = Gamma0 P_R : True
E_L, E_R isometries: E^dag E = 1_8, E_L^dag E_R = 0, E_L E_L^dag = P_L, E_R E_R^dag = P_R : True
chi E_L = -E_L, chi E_R = E_R, W unitary : True
W^dag h(p) W = [[-sigma.p, m],[m, sigma.p]] : True
sigma_i Hermitian, {sigma_i,sigma_j} = 2 delta : True
chiral blocks: E_L^dag beta E_L = 0, E_L^dag beta E_R = 1, E_L^dag alpha E_L = -sigma, E_R^dag
    ...alpha E_R = sigma, E_L^dag alpha E_R = 0 : True
E_L^dag Gamma0 (i Dslash - m) psi = (i d0 + tauO.d) phi_L - m phi_R : True
E_R^dag Gamma0 (i Dslash - m) psi = (i d0 - tauO.d) phi_R - m phi_L : True
(i d0 - tauO.d)(i d0 + tauO.d) f = -box f (8 components) : True
psi = E_L phi_L + E_R phi_R : True
chiral form of the bilinears (beta i chi, chi, alpha chi) : True
{psi_a(t,x), psi_b^dag(t,y)} = delta_ab (1/V) sum_q e^{iq(x-y)} (5 momenta, abstract mode
    ...algebra) : True
{psi_a(t,x), psi_b(t,y)} = 0 : True
the three U_k solve the Dirac equation (m = 1) : True
Grassmann components anticommute: psi_a psi_b + psi_b psi_a = 0, psi_a^2 = 0 : True
current conservation d_A (psibar Gamma^A psi) = 0 (Grassmann field) : True
dL_int/dpsibar_a = -[gS(psibar psi) psi + gP (psibar i chi psi) i chi psi + gV (psibar G_A psi)
    ...G^A psi + gA (psibar G_A chi psi) G^A chi psi]_a : True
h_KS Hermitian for real S, P, V_A, A_A : True
Gamma0 [pslash(eps,p) - m - S - i chi P - Gamma^A (V_A + chi A_A)] = eps - h_KS(p)  (plane wave
    ...e^{-i(eps t - p.x)}) : True
W^dag h_KS W = [[V^L_0 - sigma.(p+V^L), M],[M^*, V^R_0 + sigma.(p+V^R)]], V^L = V - A, V^R = V
    ...+ A, M = m + S + iP : True
Hartree potentials: dE_H/dphi*_a = (v_KS phi)_a with S = gS rhoS, P = gP rhoP, V_A = gV j_A,
    ...A_A = gA j5_A : True
```

```
densities real: Gamma0 O Hermitian for O = 1, i chi, Gamma^A, Gamma^A chi : True
chiral densities: rhoS = kappa + kappa^*, rhoP = i(kappa - kappa^*), j = jL + jR, j5 = jR - jL
    ...: True
E_H = gS/2 (k+k*)^2 - gP/2 (k-k*)^2 + (gV+gA)/2 (jL.jL + jR.jR) + (gV-gA) jL.jR : True
V^L_A = V_A - A_A, V^R_A = V_A + A_A (Hartree, chiral currents) : True
M - m = dE_H/dkappa = gS(k+k*) - gP(k-k*) = S + iP : True
dE_H/dphi_L* = (V^L_0 - sigma_i V^L_i) phi_L + (M - m) phi_R ; dE_H/dphi_R* = (V^R_0 + sigma_i
    ...V^R_i) phi_R + (M - m)^* phi_L : True
U_theta = cos(theta) - sin(theta) omega is real orthogonal; U E_L = e^{-i theta} E_L, U E_R =
    ...e^{i theta} E_R : True
U^dag beta U = beta (cos 2theta + i sin 2theta chi);  vector and axial currents invariant : True
E_H(e^{i theta chi} psi) - E_H(psi) = (gS-gP)/2 [(e^{4i theta}-1) k^2 + (e^{-4i theta}-1) k*^2]
    ...->  U(1)_A symmetric iff gP = gS : True
256 products Gamma_K are orthogonal: Gamma_K^-1 = Gamma_K^T : True
tr(Gamma_K^-1 Gamma_L) = 16 delta_KL (256 x 256) : True
Fierz completeness: sum_K (Gamma_K)_ab (Gamma_K^-1)_cd = 16 delta_ad delta_cb : True
delta E_x = tr[Sigma_x delta R] (random R, dR), Sigma_x = -gS R - gP (i chi) R (i chi) - gV G_A
    ...R G^A - gA G_A chi R G^A chi : True
E_x = (1/2) tr[Sigma_x R] : True
E_H = (gS/2)(tr R)^2 + ... = sum_c (g_c/2) tr(O_c R) tr(O^c R) and dE_H = tr[Sigma_H dR] : True
Fierz: tr[O1 R O2 R] = (1/16) sum_K tr(Gamma_K^-1 R) tr(O1 Gamma_K O2 R) for all channel
    ...vertices : True
uniform R: tr R = rho_S, tr(Gamma^0 R) = rho : True
(1/2) sum_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) = (E Gamma_0 + m)/(2E) : True
E_x(uniform) = -gS/32 (rS^2+rho^2) + gP/32 (rS^2-rho^2) - gV/32 (8rS^2-6rho^2) + gA/32
    ...(8rS^2+6rho^2) : True
Sigma_x(uniform) = S_x + V_x Gamma_0, S_x = rS/16(-gS+gP-8gV+8gA), V_x = rho/16(-gS-gP+6gV+6gA)
    ...: True
S_x = dE_x/drho_S, V_x = dE_x/drho : True
uniform Hartree-Fock = Hartree with G_S = gS + (-gS+gP-8gV+8gA)/16, G_V = gV +
    ...(-gS-gP+6gV+6gA)/16 : True
uniform matter: Hartree terms of P and A vanish (rho_P = 0, j5 = 0, j^i = 0) : True
Wick (3 particles, 16 modes): <:(psibar O psi)(psibar O' psi):> = tr(O R) tr(O' R) - tr(O R O'
    ...R) for all 18 vertex pairs : True
c7 = 8 |S^6|/(2pi)^7 = 1/(15 pi^4) and rho = c7 kF^7/7 = kF^7/(105 pi^4) : True
NUM Hartree saturation kF0 : 0.75663615070149289469
NUM Hartree rho0 : 0.00001388100213231687719 7
NUM Hartree m* : 0.34809734014624326514
NUM Hartree rhoS : 6.5190265985375673486e-6
NUM Hartree V0 : 0.13881002132316877197
NUM Hartree E/A - m : -0.028321427964738575854
NUM Hartree eps_F : 0.97167857203526142415
Hartree gap: m* - m - gS rhoS(m*) = 0 : True
m* is a stationary point of eps(m*) = ekin(m*) - (m*-m)^2/(2gS) + gV rho^2/2 : True
total energy = sum eps_k - E_H (double counting) = ekin - (m*-m) rhoS + E_H : True
Hugenholtz-Van Hove at saturation: E/A = eps_F = d eps/d rho : True
d(eps/rho)/dkF = 0 at kF0 (minimum: E/A(0.95 kF0), E/A(1.05 kF0) larger) : True
Hugenholtz-Van Hove mu = d eps/d rho = eps_F at kF = 1 (not saturated) : True
NUM Hartree kF=1: m* : 0.080964430277606126329
NUM Hartree kF=1: E/A - m : 0.41086484863433217115
NUM Hartree kF=1: eps_F : 1.9809848613081542876
NUM HF couplings G_S, G_V : -98750.0, 20000.0
HF self-consistency with the exchange self-energy: m* - m = gS rhoS + S_x, V0 = gV rho + V_x :
    ...True
HF energy: ekin - (m*-m) rhoS + E_H + E_x = eps(G_S, G_V) : True
HF Hugenholtz-Van Hove: d eps/d rho = eps_F at kF = 3/4 : True
NUM HF kF=3/4: m* : 0.36643630198154392124
NUM HF kF=3/4: V0 : 0.2610177547316155257
NUM HF kF=3/4: E/A - m : 0.039146338016119861526
NUM Hartree kF=3/4: m* : 0.36297394764803019822
NUM Hartree kF=3/4: E/A - m : -0.028065375846872960879
J(0) = Lambda^6/(720 pi^4), g_c = -1/(8 J(0)) = -90 pi^4/Lambda^6 : True
NUM NJL g_c : -8766.8181930602193513
NUM NJL g = 6/5 g_c : -10520.181831672263222
NUM NJL |M| (m = 0) : 0.55369385182924169122
```

```
NUM NJL kappa = -4 M J(|M|) (M real) : -0.0000263157928583648742
NUM NJL E(|M|) - E(0) : -1.3102449629010022473e-6
NJL m = 0: gap 1 + 8 g J(|M|) = 0 solved, M = 2 g kappa^*, stationary and lower than M = 0 :
    ...True
NJL m = 0: E(|M| e^{i theta}) independent of theta (degenerate chiral vacua) : True
NUM NJL M (m = 1/100) : 0.58531284674622051123
NJL m = 1/100: M = m - 8 g J(M) M real, stationary; E(M e^{i theta}) > E(M) for theta != 0 :
    ...True
sea at one momentum: eigenvalues -E (x8), +E (x8); kappa_p = tr[(Pi_-)_{RL}] = -4 M^*/E;
    ...tr[Pi_- h_0] = -8E + 8 Re[(M-m) M^*]/E : True
boolean checks: 90   failed: 0
VERDICT: ALL CHECKS PASSED
```

4.4 Hashes of the files

file	SHA-256
7+1.nb (evaluated)	1314b1e09e6b6eefbd9ee5974e19d186a5e996907b2eb68ddb6f23aae85aaaa5
4+4.nb	740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5
7+1_2026-10-26_before_dirac16complex.nb (in backups/, workspace only)	f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce
tools/cayley_dickson_formulas.py	bdde8c4c8199b09ef36ae4e15d5d53bc62744fc85b4c5ba52eab5516f77b5d5f
tools/cayley_dickson_match.py	73f2cc584bbe1a3620c50890ffe5fb299857b04a14bc34fed2f9eaefd2e36c4e
tools/classify_clifford.py	42291276f46b046183fdbb7b62be7c5cedfd695fe9da96ac256afa23580930a9
tools/doc_common.py	1e37b140f8334aa813418f4a79b7d0a8e67740b2c4a3fe4d5f4e34abde690d3e
tools/doc_coupled.py	7ffc2ed938d400ca7579b88f2963864dfcd55a800744502d6fcf1a64e1ed52f6
tools/doc_coupled_provenance.py	0336f71a9e6a042b5c916cbef255ca829eaa087f6bd80079040a67778cd57a10
tools/doc_coupled_symbols.py	849714a2b35c06c8d69464e1535cde3e60cf8866448782c5659a66424562719f
tools/doc_dirac16.py	3f81df5fe2237843edd29116e5676679527b80e946623890e6289a1487f754f1
tools/doc_dirac16_common.py	9324b4fae59e51019125ebe09157f3c8c9489560e1383a697fe6c22a3a93e208
tools/doc_dirac16_provenance.py	daf12ff5757d0c4ca8857840d027907f21429e7340b2f02c3b50fbf8edb2d663
tools/doc_dirac16_symbols.py	69224cd370206759e31abd225fc556630455d6d99226db0bc46068eae9b27db0
tools/doc_gravity.py	184c7aa38742a2057449a98037a8521842fc81072613611c228c63c17083af0c
tools/doc_gravity_provenance.py	3845317cdef374106b60524416eda8e3a81fd5d2a8e35c327c3f1ae32925b0f4
tools/doc_gravity_symbols.py	fcff3b16ea23fa62fe205e66310552f7d3e119c619c150379e67bd17150b5d22
tools/doc_ks_gravity.py	9aa73527f90318ee3444fcf97af41d2255a5d8fb1697d9fa0128ca671e94e51f
tools/doc_ks_gravity_provenance.py	112b51528ab67acd8ed110b57c347352c602c984eae1408d7dd36e41c74fc0ea
tools/doc_ks_gravity_symbols.py	2acdf7f901b9cb7c3047b6de6895d32385be37d16789df9d9124f976c6d5a515
tools/doc_provenance.py	12a523b5d7ebbc2f6a064fcc4d4e5c9fde37cd8dcfcf2eaf798da4461ff86b84
tools/doc_readme.py	9d876e736c37a2b40464401c68d4bccc9c01287230fbf29bb895a47ec4493756
tools/doc_readme_stage3.py	9d52a48e1d9d20a19b0874f144cf5bfe5204a81a3c0f448ef44f510f48a1cf91
tools/doc_readme_stage4.py	9eaaf81a92774f198f73aab99a8f4a3f4d6bccf8ff815441f8a07f6e5b76cb46
tools/doc_sm.py	d7245d8c5f5dfd2687e987d43d9d544d0a1a46538901066e244fe0261bd427c3
tools/doc_sm_provenance.py	742dec7e9ea7214a8921777a260838830c18204223b5cbb37b57962ab522d01c
tools/doc_sm_symbols.py	a019dcbf5e68f3f00f41613f5c592b84eafa8b48d4023208516fd5d93b8893f8
tools/doc_stage3_common.py	af753eafd8fda8b39fd52674aea080177aa53e6be18e331c4e3a0cd872c54e31
tools/doc_symbols.py	86f75c43deb747dbb9c05c86a7775040a8b371a626eee1041f7283a0c8f6ac7f
tools/docgen.py	5db6f40cb1eb10e48da5b69c3893829737c3d8d2e90edcb28311a980c1aa5fea
tools/eval_cells_log.wl	833b6bd854cab253e5bb935c28f28ab50b286f409d130f433fa5dfb9b9236bbe
tools/extra71.py	9935ef99cd9300f2a7d4f06d36cf7e173912ebae61b6feff448ebe0588853168
tools/git_push.wl	ac05a6d0256834b48cc9fa943fa247bccc574f723e68f053d00bfba9aca2e21a
tools/lovelock71.py	d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af95a1
tools/make_7p1_nb.py	354a0210e60510499753c2cf91274850cedef502d6f705efc410c5a8bbae7b2d
tools/make_docs.py	5f2fedb05c0761e6f91c23e418350bf6370ec9b9466b6111b39cd8a9b7950cca
tools/nb_part_coupled.py	49c61aeb6affe14ebbcccb32d2069504b9fc035313ea2fabbd0a3392a96133f5
tools/nb_part_dark.py	f3a745dd293c22c81ae44e1a1fb5aa88aeff3f5f697524ddf1d7a7a1f9901c02
tools/nb_part_gravity.py	c4caf2ad5c932e8f603341edade5f4b98fa784feb05906c051eab7c884cd68f8
tools/nb_part_ks_gravity.py	a656c0ad040b4576b75b17e029545109e531c8e656ad99bfdb5cba946e967772
tools/nb_part_sm.py	43e92beed57453a4b57931de6d603ac6c218efba6e4005133b69b8054a975c5b
tools/nbparse.py	ad2255a8864be7447a55023840b327f789cb8c9d868d42261b19261653847423
tools/nbwrite.py	dcbe8ce5b3203c89a1d17fbe946e1b6b7bb7f8895afa944743d651cfefadeffb
tools/octalg.py	36b50120b69a963e252e07d60ec007755220822b4ad339716ea7aed93d58eb24
tools/ref44.py	ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064faf5

file	SHA-256
tools/ref71.py	1a7a06eb0add3f459611776dc9c474a5bb959d947a9db949db4612b1ab9434e0
tools/ref_coupled.py	695dbd9375753653472d9c37502f0067c4a64fe01a64ba8c05e8b2c5da85aa58
tools/ref_dark.py	36de56fe4ebdf86bd38c927b0c274fccce7cab9b4948d0db73a333f75fea0b37
tools/ref_dirac16.py	32e8c50de1c480bc4ef50d254d04a9aaa0e56b486198edf0355905c0dda25674
tools/ref_gravity.py	f07cf8adaff585af89c87d19532f1aaa7db606ad88ae90483b2d1f41341edf36
tools/ref_ks_gravity.py	b6488ef24b1c16aacd8ff6cf6a5a1ba3f316e082449ad97480022d1a2638327f
tools/ref_sm.py	f3f80806893f42bbf166e1f1de896ffd611dbfe847685da8e3ee367e9cde6aba
tools/run_eval.wl	85ba54f18a8362a69c991e66a207230dd001ba18e52639fc1edf7cac761aac30
tools/verify_7p1_nb.py	86558818090d65cb821115647a762b4dd2d915a8be13213254811c0eba232e8e
tools/watch_requests.wl	d9ceedb0b45ac3cf9b4dd8519ac164953f2c80e5188b1557fe20ef5107ae352d

5. Process and environment

- Workspace: `D:\Developer\github\7+1` on Windows. Python and LaTeX ran in a Linux shell with that folder mounted; intermediate files went to `build/`, which the repository's .gitignore excludes.
- The notebook is generated by `tools/make_7p1_nb.py` (sections 0–6 by the code of the first stage, unchanged except for the signature explanation of 27 October 2026; sections 7–10 appended), with cells written by `tools/nbwrite.py` as pure ASCII (named characters such as `\[CapitalGamma]`, and `\:03d6` for ϖ).
- All Wolfram Language work ran in the Wolfram 15.0.1 front end (the program of section 4.1), version 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026). Jobs were requested without keyboard input through `tools/watch_requests.wl`: loaded once in that front end, it reads `build/request.txt` every 10 seconds and evaluates `tools/run_eval.wl` (job `eval`), `tools/git_push.wl` (job `push`) or a development file such as `tools/eval_cells_log.wl` (job `run`).
- Python 3.13.17 with sympy 1.14.0, mpmath 1.3.0 and numpy 2.5.3; fontTools reads the font coverage for the LaTeX output.
- This record and the main document and symbol dictionary of this stage are generated by `tools/make_docs.py` from the modules `docgen.py`, `doc_common.py`, `doc_dirac16_common.py`, `doc_dirac16.py`, `doc_dirac16_symbols.py` and `doc_dirac16_provenance.py` of the folder tools/; one source renders Markdown and LaTeX, and every matrix, check list and number is computed by Python or parsed from the evaluated 7+1.nb. The LaTeX is compiled with XeLaTeX (TeX Live 2022, `latexmk -xelatex`, fonts Latin Modern and DejaVu).
- Pushes: `tools/git_push.wl` runs the user's Git for Windows from the Wolfram front end with `GIT_TERMINAL_PROMPT=0` and `credential.interactive=never` (no login prompt), pushes the branch main to origin `https://github.com/once-ere/7plus1.git`, and logs `git ls-remote origin`; the published commit is compared with the local one after every push.
- Order of the work: backup of 7+1.nb; Python reference `tools/ref_dirac16.py`; notebook sections 7–10, developed cell by cell; evaluation and cross-check (`tools/verify_7p1_nb.py`); documentation. The notation of the complex scalar source was changed from μ to ϖ after the first evaluation, and 7+1.nb was regenerated and evaluated again; the boost, the proof matrices and Euler's Gamma function were renamed in the documents only ($\mathcal{B}$, Y, Z, Γ_{E}).

6. Authorship

Notebook sections 7–10, tools and documents written by Claude (Anthropic, model claude-opus-5-5) for Nash, completed on 2026-10-27, from the task described in section 1 and the first part of 7+1.nb described in section 2.1.